From the Ivory Tower to Capitol Hill: Gender, Mobility, and Expert Advisory Work

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Abstract

Examining academic economists' engagement with public policymakers, this paper investigates whether geographic mobility friction contributes to gender disparities in expert testimony before the U.S. Congress. We argue that expert engagement is a spatially concentrated activity and that women face systematically higher costs of geographic mobility, constituting an overlooked supply-side constraint on their participation in high-impact expert advisory work. Using a COVID-19 related policy shift from in-person to online hearings as a source of exogenous variation in mobility requirements, we show that reducing geographic friction significantly changed the composition of expert witnesses. Constructing a dataset of the universe of economists who have testified since 1969 and a large at-risk group of intellectually proximate economists, we find a consistent female disadvantage in the odds of appearing as a witness. The geographical distance from Washington DC reduces the odds of testifying for all economists, but has a stronger effect on women. Following the shift to online hearings, female economists were significantly more likely to testify relative to men. Geographic mobility creates value for organizations and individuals, but mobility friction falls disproportionately on certain groups. Our results suggest that reducing mobility friction can help capture this value even for episodic opportunities that require travel. Working-from-anywhere policies, by lowering this friction, can broaden organizations' access to a wider pool of experts.

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1 Introduction

Experts provide organizations and policymakers with specialized knowledge through formal advisory appointments, scientific advisory board service, consulting engagements, and public testimony (Perkmann et al., 2021, 2013). Yet scientific expertise does not translate equally into advisory influence. Male academics are nearly twice as likely as comparably qualified women to serve on corporate scientific advisory boards (Ding et al., 2013), a pattern echoed in university-industry collaboration (Tartari and Salter, 2015), patenting (Ding et al., 2006b), and academic entrepreneurship (Abreu and Grinevich, 2017). Beyond raising concerns about equity within academia, limiting advisory appointments to a narrower pool of experts implies a misallocation of talent (Hsieh et al., 2019), and shapes which problems receive attention and whose interests are served in organizational decision-making (Einiö et al., 2023; Koning et al., 2021; Nielsen et al., 2018). Existing accounts of this gap emphasize individual choices or gatekeepers' preferences (e.g., Ding et al., 2013; Tartari and Salter, 2015), but neither considers a structural feature of advisory work itself: unlike knowledge that travels through publications, exercising advisory influence typically requires physical presence in a small number of decision-making hubs, and the resulting *geographic mobility frictions* (Choudhury, 2022) are well documented to fall unevenly across individuals. This paper asks whether these frictions, rather than differences in preferences or gatekeeping, helps explain who gains access to advisory influence.

Many forms of expert advisory work are geographically concentrated, requiring travel to Silicon Valley, Washington DC, corporate headquarters, and conferences, and prior research shows that geographic proximity facilitates knowledge transfer and collaboration, while reducing mobility frictions expands collaborative activity (Catalini et al., 2020; Ding et al., 2010; Jaffe et al., 1992; Katz, 1994). These costs are unlikely to be borne equally. Women continue to shoulder a disproportionate share of caregiving responsibilities, making travel more costly and contributing to sorting into jobs that offer greater work-life balance (Barbulescu and Bidwell, 2013), stronger preferences for temporal flexibility (Goldin, 2014), shorter commuting distances (Barbanchon et al., 2021), and greater sensitivity to geographic distance in the academic labor market (Bisantis, 2026). What is less clear is whether these constraints extend to episodic professional opportunities, where the time-bounded and infrequent nature of the commitment may make temporary substitution of caregiving responsibilities more feasible. We therefore argue that geographic mobility friction constitutes a potentially important, but largely overlooked, supply-side constraint on women's participation in expert advisory work.

To test whether this friction shapes access to advisory influence, and contributes to gender disparities in who exercises it, we study university researchers' testimony before the U.S. Congress. We do so as this setting combines a consequential form of expert advisory work with unusual opportunities for observation and causal identification. Hearings provide a comprehensive public record of expert participation (Ban et al., 2023) and underwent an abrupt transition from exclusively in-person to virtual hearings during the COVID-19 pandemic, an exogenous reduction in geographic mobility friction that lets us compare participation before and after the requirement of physical presence was removed.

This strategy, however, depends on knowing not only who testified, but who else could plausibly have done so. Studying selection into expert advisory work presents an important empirical challenge because the relevant opportunity set is rarely observed. While many forms of expert engagement record who ultimately participates, organizations rarely document the broader set of experts who were eligible or considered for participation. We address this challenge by developing a novel approach to approximating the opportunity set in settings where individuals leave a public record of their expertise, such as academic scientists through publications. For each congressional hearing, we construct a hearing-specific risk set using only information available at the time of the hearing, combining observable characteristics with the semantic similarity of scholars' published research to identify academics with comparable expertise to the invited witness. This approach enables the study of selection into episodic expert advisory roles by recovering an otherwise unobservable opportunity set and is applicable to contexts in which expertise can be inferred from an observable knowledge base.

We construct a novel dataset capturing the universe of academic economists who testified before Congress between 1969 and 2023, comprising 5,331 testimonies by 2,036 unique individuals. Combining these data with our hearing-specific opportunity sets allows us to estimate selection into congressional testimony conditional on being a qualified candidate, controlling for hearing-level characteristics. Across all specifications, we document a persistent gender gap: conditional on productivity, seniority, institutional prestige, professional networks, and geographic location, women have 13–16% lower log odds to testify before Congress than comparable men. Geographic distance from Washington DC reduces testimony propensity for all academics, but the penalty is significantly larger for women. Consistent with our theoretical argument, the transition to virtual hearings reversed this pattern. Women became significantly more likely than comparable men to testify, with the effect concentrated among those located furthest from Washington, D.C., the group least likely to testify under in-person hearings and most likely to do so once mobility constraints were relaxed. Together, these findings identify geographic mobility friction as an important and previously overlooked supply-side mechanism underlying gender gaps in access to expert advisory work.

First, we contribute to the literature on geography and knowledge work, which has shown that geographic frictions shape collaboration, knowledge transfer, and innovation (e.g., Jaffe et al., 1992; Katz, 1994; Ding et al., 2010; Catalini et al., 2020). Existing evidence, however, paints an incomplete picture. While reducing communication costs disproportionately benefits women scientists' collaborations and productivity (Ding et al., 2010), reducing travel costs appears to benefit primarily men (Catalini et al., 2020). We help explain these findings by studying expert advisory work, a setting in which communication can occur remotely but participation has traditionally required physical presence. Our findings suggest that an important dimension of geographic friction is not the cost of travel alone but the organizational requirement for in-person participation, highlighting how the design of knowledge work shapes who can participate in it.

Second, we contribute to the academic engagement literature by advancing understanding of expert advisory work as an important yet relatively understudied form of engagement between academics and organizations (Perkmann et al., 2013, 2021). We document substantial gender disparities in participation in these influential advisory roles and identify geographic mobility frictions as an important structural mechanism underlying these disparities. Finally, we show that communication technologies that reduce the need for physical presence can mitigate these frictions and increase the participation of women in expert advisory work. Together, these findings extend existing accounts of academic engagement by highlighting how the organization of advisory work shapes who participates in it.

Third, we contribute to the literature on geographic mobility and gender inequality in labor markets, which has shown that differences in commuting, relocation, and workplace flexibility shape occupational sorting, career progression, and earnings (Blackburn, 2010; Goldin, 2014; Mincer, 1978; Peterson, 1979; Sandell, 1977; Schoenbaum, 2017; Sorenson and Dahl, 2016; Widiss, 2012; Borghorst et al., 2024; Casado-Díaz et al., 2023; Barbanchon et al., 2021; He et al., 2021; Wiswall and Zafar, 2018). We extend this literature to episodic professional opportunities that require temporary travel rather than permanent or long-term relocation. Our findings show that geographic mobility frictions influence access to prestigious professional opportunities even when mobility demands are infrequent and short-lived, suggesting that the consequences of mobility constraints extend beyond employment choices to the allocation of influential advisory roles.

The rest of this manuscript is organized as follows. Section 2 outlines the background literature on gender disparities in academic engagement and geographic mobility, from which hypotheses are derived. In section 3 congressional hearings as an empirical context is described and in section 4 and section 5 our data, variables, and methods are outlined. In section 6 our predictions are tested on the universe of academic economists who have advised members of Congress via hearing testimony from 1969-2023 with a constructed at-risk population of intellectually close academics. Finally, implications, limitations, and future work are discussed in section 7 and section 8.

2 Background

Our hypotheses regarding the role of geographic mobility friction in shaping gender disparities in expert advisory work build on two streams of literature. Empirical research has documented persistent gender disparities in academics' engagement with audiences beyond the university, and a separate body of work has examined gender differences in academics' geographic mobility over the course of their careers. The review that follows offers a review of this prior research and situates the present study's mechanism and hypotheses within it.

2.1 Gender and Academic Engagement

University researchers play a crucial role in translating research into innovation and public policy (Bikard and Marx, 2020; Kogan et al., 2017; Sorenson et al., 2006; Sorenson and Fleming, 2004), and a large literature on academic engagement has documented the individual-level determinants of this external knowledge transfer across modes ranging from university-industry collaboration and technology licensing to academic entrepreneurship and advisory roles (Abramo et al., 2011; Ding and Choi, 2011; Ding et al., 2013; Lawson, 2022; Lawson and Salter, 2023; Perkmann et al., 2013, 2021; Tartari and Salter, 2015). This literature has centered mainly on commercial, STEM-oriented engagement, leaving academics' engagement with public and non-profit audiences comparatively understudied (Lawson, 2022, p. 148).

Within this literature, gender has emerged as a consistent predictor of engagement. Participation in commercial engagement modes is highly stratified by gender, even among otherwise similar faculty (Abreu and Grinevich, 2017; Ding et al., 2006b; Murray and Graham, 2007; Perkmann et al., 2013, 2021; Tartari and Salter, 2015). Scholars have offered two broad explanations for this pattern. Supply-side accounts point to differences in field sorting, professional networks, and willingness to engage publicly (Burrage et al., 2025; Ding et al., 2013; Hunt et al., 2013; Murray and Graham, 2007; Sarsons and Xu, 2021; Sievertsen and Smith, 2024), while demand-side accounts emphasize the role of gatekeepers and gender stereotyping in expert selection (Coil et al., 2024; Ding et al., 2013; Tartari and Salter, 2015). Missing from both perspectives is the possibility that expert engagement is a spatially concentrated activity, and that women face systematically higher costs of geographic mobility. We therefore expect a baseline gender gap in participation in expert advisory work:

Hypothesis 1. *Female academics are less likely than comparable male academics to appear as expert witnesses before Congress.*

2.2 Gender and Geographic Mobility

Geographic mobility is a key channel through which high-skilled workers build networks, transfer knowledge, and create value for organizations and individuals (Agrawal et al., 2006, 2008; Almeida and Kogut, 1999; Choudhury, 2022; Criscuolo, 2005; Ganguli, 2015; Jaffe et al., 1992; Moser et al., 2014; Saxenian, 2005; Singh, 2005), but mobility is costly, with relocation probability declining sharply in distance (Vaccario et al., 2021), and these costs fall disproportionately on women. In academia specifically, women's professional world tends to be less internationally mobile than men's, particularly in the natural sciences (Jöns, 2011), and this gender gap in mobility persists across fields and countries even as it has narrowed over time (Momeni et al., 2022; Zhao et al., 2023).

Qualitative and quantitative research points to caregiving responsibilities as a central mechanism behind this pattern. For decades, researchers have documented how labour market participation is strongly tied to life-cycle events such as parenthood and marriage, with especially strong effects on women (Blackburn, 2010; Goldin, 2014; Mincer, 1978; Peterson, 1979; Sandell, 1977; Schoenbaum, 2017; Sorenson and Dahl, 2016; Widiss, 2012). Within academia, caregiving obligations "stick" to the mobile academic in ways that are difficult to disentangle from mobility decisions themselves (Henderson, 2021), and academic mobility continues to be discursively constructed as essential to career success, with frequent travel considered necessary despite advances in virtual communication technologies (Cohen et al., 2020). These caregiving-related frictions are not confined to long-term relocation: early-career researchers report comparable tensions when weighing whether to pursue international mobility at all (Tzanakou, 2021), and mobility friction is similarly evident in women's shorter commute patterns and higher demand for temporal flexibility (Borghorst et al., 2024; Casado-Díaz et al., 2023; Goldin, 2014; He et al., 2021; Barbanchon et al., 2021; Wiswall and Zafar, 2018).

Taken together, this evidence indicates that gendered mobility constraints affect women's labour market decisions well beyond academia's own borders. Women face higher effective costs of geographic displacement due to disproportionate caregiving responsibilities and the asymmetric household dynamics that make them more likely to bear the costs of spatially demanding professional commitments. These constraints have been documented in job search, commuting, career mobility, and academic mobility, but their reach may extend further still. If mobility constraints shape access to labour markets and academic careers, they may similarly shape access to professional opportunities within a job that require episodic but binding geographic presence, such as episodic expert advisory roles.

Congressional testimony represents precisely this type of spatially concentrated, episodic commitment. It requires travel to Washington DC, operates on compressed and non-discretionary timelines, and cannot be rescheduled around caregiving obligations. Unlike permanent job moves, it does not alter primary employment, but it imposes temporary geographic mobility layered onto existing obligations. If the mobility constraints documented in the literature carry over to this domain, we should expect participation to decline with geographic distance from Washington DC, and this effect should be larger for women than for men. We thus expect:

Hypothesis 2. *Geographic distance from Washington DC to the academic's affiliation is negatively associated with the probability of academic participation in congressional testimony, and this effect is larger for female academics than for male academics.*

If mobility constraints drive this gender gap, then reducing the geographic friction embedded in congressional testimony should disproportionately increase women's participation. The shift to virtual hearings during the COVID-19 pandemic represents an exogenous reduction in that friction. Under the mobility constraint mechanism, this shift should increase female participation most among women for whom travel costs were highest, namely those at institutions far from Washington DC. We further expect:

Hypothesis 3. *The shift to virtual congressional hearings disproportionately increased the probability of female academic participation, with effects concentrated among academics at geographically remote institutions.*

3 Empirical Context and Strategy

Our identification strategy relies on differences in witness selection between men and women within a matched pool of plausible witnesses. If travel and mobility constraints constitute an important barrier to participation, and if these constraints disproportionately affect women economists, the transition to virtual hearings should increase geographically distant women's odds of testifying, relative to men.

3.1 Policymakers Reliance on Science

Science has long been regarded a crucial source of evidence and expert knowledge that informs policy issues (Prewitt et al., 2012). Amidst pressing societal challenges like public health crises, climate change, and rapidly evolving technologies, scientific advancement has an intricate link with the policy process (Azoulay and Jones, 2020).

Academics have a long history of engaging with the U.S. government, to inform political actors on a broad range of issues (Weingart, 1999, e.g.,). For a given academic, this engagement can take various forms, such as serving on boards and committees (Ding et al., 2006a, e.g.,), occupying political positions (Hirschman and Berman, 2014, e.g.,), or providing expert testimony and advise (Ban et al., 2023; Maher et al., 2020). More recently, the push for government towards "mission-oriented approaches to science and innovation policy has required policymakers to engage more intensely with scientific research" (Beck et al., 2022, pp. 147–148).

The empirical setting of this study is expert testimony before committees of the U.S. Congress. Congress serves as a good setting and proxy for policy engagement for various reasons. First, Congress has a long history of convening hearings with in-person witness testimony, serving as a consistent baseline for our study. Second, congressional hearings are meticulously documented and digitized, allowing for the extensive and rich primary data collection undertaken in this study. Third, committee hearings are considered one of the main settings in which members of congress collect and assess information before congressional deliberation, making it a high-impact arena for academic engagement, one where representation might matter to consequential outcomes (Ban et al., 2023). Finally, Congressional hearings represent a highly visible and influential setting for academics to share their expertise, which differs from contexts where the impact of removing geographic friction on women's participation has previously been studied (Biermann, 2024).

3.2 Academics in Congressional Hearings

The U.S. Congress holds substantial authority as granted by the Constitution. All legislative powers within the federal government reside in Congress, which has jurisdiction over budgetary appropriations, oversight, and investigative hearings (Cushman, 2014). The House of Representatives and the Senate are organized into committees, which are specialized sub-groups, staffed by elected officials from both the majority and minority party. Committees have jurisdiction over particular issues, providing expertise, information, and recommendations to their respective chambers.

Members of Congress contribute to making crucially important decisions, and navigate complex policy choices, often in areas they lack expertise in, and under time constraints (Ban et al., 2023). To develop their expertise, members rely on information provided by witnesses from the public, interest groups, federal agencies, and academia, obtained through congressional hearings. Witnesses submit written testimony in advance and deliver oral testimony at the hearing, receiving questions from members of the committee following their statement.

Systematic empirical work on Congress' use of academic witnesses is scarce but finds that under conditions of uncertainty, complexity, or ambiguity, expert witnesses tend to be used more (Ban et al., 2023). In fact, researchers have found that increasing complexity and uncertainty in the political landscape has led policymakers to rely more heavily on expert communities over time (Ban et al., 2023; Furnas et al., 2024; Jacobs and Page, 2005). Turning to the composition of witnesses, evidence is scarcer. Coil et al. (2024) focused on a sample of Senate and House committees and found that during the 110th through 116th Congresses (2007-2021), women made up just 24.5 percent of witnesses. In fact, 40 percent of witness panels in their data were men-only, contrasted to just 1.1 percent of women-only panels.

This past literature highlights that while members of Congress increasingly rely on expert witnesses, disparities between demographic groups in witness representation are a concern, both for the opportunity of the individual, for the quality and breadth of knowledge accessed by Congress, and for the representation of diverse views and interests.

3.3 COVID-19 and the Introduction of Virtual Hearings

A central challenge of studying the role of mobility constraints in academic engagement with policy advising are endogeneity concerns. The first and most prominent potential source of endogeneity is selection. Economists who testify are positively selected on unobservables correlated with both gender and perceived mobility costs, so that women who reach the witness pool despite structural barriers may be more prominent or more mobile than the average female economist, biasing naive comparisons toward underestimating the true mobility penalty. Second, women may select into institutions at systematically different distances from Washington for reasons correlated with their propensity to engage in advisory work, e.g., caregiving constraints that simultaneously shape residential location choices and professional commitments, meaning the association between gender, distance, and testimony propensity across individuals may partly reflect sorting into location rather than the causal effect of distance itself. We address these concerns by exploiting the shift to virtual hearings induced by the COVID-19 pandemic as an exogenous source of variation in mobility costs, a change driven by public health policy rather than by any individual academic's prior engagement, network position, or location choice, and one that reduced effective travel costs differentially for those at greater geographic distance from Washington.

The WHO declared COVID-19 a global pandemic on March 11, 2020, and in the time that followed, organizations and institutions were forced to change their daily operations dramatically (WHO, 2020). Due to lockdowns and quarantine measures, communication was increasingly facilitated online. In Congress, policymakers adopted new rules of congressional proceedings to keep the legislative body of the country running. The US Senate held its first virtual hearing on May 5th, 2020, and the US House of Representatives adopted rules allowing remote committee hearings on May 15th, 2020 (Jones, 2020). Since January 2023, Congress has mostly returned to in-person hearings but do allow for some hybrid formats.

This change meant that witnesses now could give testimony from wherever. Prior to COVID, all hearings took place in person, meaning that invited witnesses had to travel to Washington D.C., usually at their own expense, to give testimony. We leverage this exogenous technological change to estimate the causal effect of reduced mobility friction on the probability of women to testify.

On the demand side, we assume that controlling for hearing topic invitations for women to testify remain stable. We know that women sort into particular subfields in economics, and so any change in the composition of expert witnesses may be driven by the change in demand for expertise on different policy issues, e.g., health economics. We address this concern by estimating a conditional logit, where identification comes from within hearing variation only, taking out hearing-level characteristics, such as the topic. On the supply side, we assume that women are more likely to accept an invitation because of the online format. For geographically distant researchers, costs associated with going to Congress are reduced, and so this highly visible, high-impact dissemination opportunity might be more attainable.

4 Data

This paper draws on multiple data sources to assemble a dataset capturing the universe of academic economists, who have testified before the U.S. Congress from 1969 to 2023, and a large at-risk group of plausible alternative witnesses. To build our dataset of congressional testimonies, we first extended the work of Maher et al. (2020) with detailed records from 2017 to 2023. Our at-risk group is constructed from the authors of papers semantically similar to each focal witness' stock of publications prior to the hearing. We then supplemented this data with publication output (from Open Alex), alt-metric indicators (from Dimension), and geographic distance to Washington DC (from Google Maps), to build a career panel.

The selection of economists as the focal population for this research is deliberate and methodologically driven. Economics is characterized by a remarkably stable and entrenched hierarchical structure, with minimal disruption to its institutional ecosystem over time (Fourcade et al., 2015). This stability is evident in the persistent dominance of a small set of elite academic institutions (Korom, 2020), the consistent prestige of top five journals (Heckman and Moktan, 2020), and the self-perpetuating networks of academic and professional recognition. The field operates through tightly-knit "elite clubs" – networks of institutions, research centres, and professional associations that exercise significant gatekeeping functions, making the boundaries and key actors relatively well-defined and traceable (Carrell et al., 2024; Freeman et al., 2024). Moreover, there is evidence to support the consideration of economics as a discipline with a strong policy orientation, with economists frequently serving as key advisors, consultants, and expert witnesses in policy formulation across various domains of governance (Hirschman and Berman, 2014; Maher et al., 2020).

From a methodological standpoint, there are three particular advantages of focusing on economists. First, the pandemic not only forced organizations to introduce technological tools that facilitated remote work, but also disrupted researchers' established routines. The decrease in time devoted to research during the pandemic could affect our results, if it frees up the researcher's time to devote to other tasks, such as testifying before Congress. Studying economists is useful since the bench sciences who rely on lab equipment to conduct their research, such as biochemistry and biology, reported a much larger disruption to their work compared to mathematics, statistics, computer science and economics (Myers et al., 2020). Second, the JEL code classifications provide a standardized taxonomy we can map onto the hearings, thus controlling for granular hearing topics. Finally, a methodological advantage of focusing on economists is the ability to construct a well-defined at-risk population. By finding similar scholars to the focal witnesses, we can identify the set of scholars whose expertise was substantively relevant to a given hearing at the time it took place.

4.1 Congressional Hearing Witnesses

4.1.1 Congress data from 1946-2016

To build our dataset of congressional witnesses, we began with data provided by Maher et al. (2020). This dataset encompasses testimonies from social scientists appearing before U.S. congressional committees from 1946 to 2016, detailing information on witnesses and their affiliations, the committees involved, and the hearing topics. In total, the dataset comprises 23,016 testimonies by 7,295 unique individuals. Economists constitute the majority of social scientists testifying: 15,050 testimonies, or 65 percent of the hearings recorded by the researchers, were delivered by economists, representing 3,883 unique individuals. We truncated this data from 1969 to avoid focusing our analysis on years where women were barely represented in the economics profession.

4.1.2 Congress data from 2017-2023

This was further expanded with a primary data collection effort to include 2017-2023, as these years are critical to our empirical design. We did so by collecting hearing records from the congressional archive GovInfo and extracting the witnesses using an API access to a Generative-AI tool developed by OpenAI (GPT-4o) (Congressional Hearings, nd). In total, 7,381 hearing transcripts as PDF files were collected.

To extract the precise information about witnesses appearing in congressional hearings, a Python script was developed, which contained three steps. In the first step, each hearing file was parsed and truncated using keywords indicating the presence of witness information, to limit the amount of text sent to GPT through the API. In the second step, a prompt containing a small sample of the desired data schema was sent to GPT-4o (see prompt in appendix A). For consistency, the extracted variables followed those in the Maher et al. (2020) dataset, with one caveat, the addition of an indicator of whether a hearing was online or in-person, as this information is crucial for our identification strategy. The third and final step entailed the organization of the output data in a CSV format.

The extraction identified 27,582 witnesses (18,602 unique) out of which the GPT-4o had identified 3,512 (2,855 unique individuals) to be academics. We then manually reviewed the 3,391 cases where GPT-4o had identified a field for the academic (e.g., professor of management) and determined which fields were within economics. For the remaining 121 witnesses where no field was inferred, we manually identified the person's field using information from their CV and university page. Removing non-economists reduced the sample to 942 economists (742 unique).

Figure 1 provides an overview of the number of academic congress witnesses by year, from different sources. The Maher et al. (2020) data is depicted with two lines, one for all social scientists and one for all economists[1] between 1946 and 2016. The most inclusive dataset is by Ban et al. (2023), which includes all think-tank and university witnesses from 1960 to 2018. Think-tank and university researchers made up 8.45 percent of the total witness pool. Next, economists in our sample are a subsample of the Maher et al. (2020) and includes all economists with a university affiliation at the time of the hearing. Finally, the red line represents all academic economists from the data we collected from 2017 to 2023. Comparing the number of economists per year between data sources, there seems to be a similar pattern. The increase in witnesses around 1970 might be explained by The Subcommittee Bill of Rights in 1974 which increased the autonomy and authority of subcommittees and gave each subcommittee distinct areas of jurisdiction. The bill incentivized committees to create more subcommittees, and to handle a broader range of issues (Rohde and Rosenthal, 1974). The subsequent decreasing trend in witnesses from the 1990's could be explained by a 1995 reform that reduced numbers of subcommittees, counteracting the effects of the Bill of Rights (Ban et al., 2023).

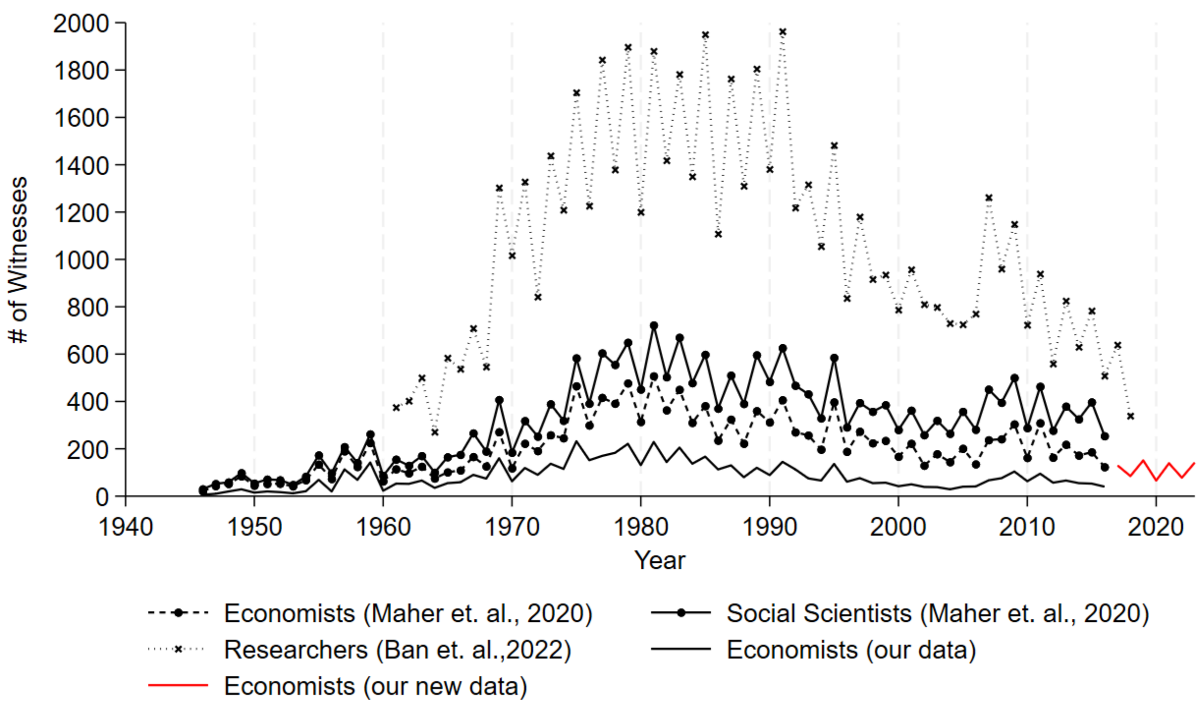


Figure 1: Number of witnesses by type and data source

[1] Not just academic economists, but also those working at think tanks, government agencies etc.

4.2 At-risk population

A central challenge in estimating gender gaps in congressional testimony is defining the population of academics who could plausibly have been invited to testify. Estimating whether women are underrepresented conditional on being a plausible candidate requires identifying who could have been called but was not. To address this, we construct an at-risk population for each witness appearance using semantic similarity matching on publication records.

For each of the 4,535 witness appearances with available publication records, we identified the witness's stock of publications in the five years prior to the hearing, averaging 13 publications per witness. We queried this publication stock through LogicMill, a tool for measuring semantic similarity between academic publications at scale, retrieving the top 500 most semantically similar publications per paper. This yielded 3,498,990 unique matched publications, with a mean cosine similarity of 0.981, confirming that retrieved papers are substantively close to the focal witness's work. To validate the use of the five-year publication window as a proxy for hearing-relevant expertise, we sampled all testimonies between 2017 and 2023, extracted the written testimony, and measured the semantic similarity between it and the witness's full publication stock, ranking publications by similarity. Appendix D reports the distribution of the year distance between each witness's most similar publication and the hearing year, separately by gender. The distribution is right-skewed and concentrated in the first decade prior to the hearing, suggesting that the most topically relevant work tends to be relatively recent, broadly consistent with the use of a five-year window. We note, however, that the distribution has a long right tail, indicating that for some witnesses the closest matching publication predates the window.

From the matched publications, we extracted the authors, yielding an initial candidate pool of 3,268,902 author records. For each witness-hearing pair, we computed each candidate's number of appearances across matched publications as a measure of proximity intensity to the focal witness. We then applied a set of filters to retain plausible candidates: we dropped the witness themselves, removed candidates with fewer than three appearances across matched publications, and required that candidates have at least one publication, at least one US affiliation, and at least one university affiliation in the five years prior to the hearing. After filtering, the mean candidate pool per hearing stands at 607, with a minimum of 3 and a maximum of 5,252. We further capped the candidate list at the top 20 ranked candidates per hearing by appearance count. Of these, 47% met both the US affiliation and academic institution criteria and were retained. This yields 21,412 unique candidate authors, for whom we retrieved full bibliographic histories from OpenAlex to construct career panels analogous to those of actual witnesses. After constructing career panels for each at risk individual, we removed outliers[2] and in the final sample, each witness-hearing appearance has

[2]For example, OpenAlex falsely attributes publications to profiles in some cases, which can distort seniority, calculated as time since first publication. We drop witness appearances and at-risk individuals where the economist has an academic age above 65.

a mean risk set of 17 individuals, median of 14, and a minimum of 5. It has been demonstrated that an at-risk to event ratio of 5:1 or higher results in little loss of efficiency in estimations (Ding et al., 2013).

4.3 Variables

We compiled data from various data sources to create covariates at the individual and hearing level. All time-variant variables were constructed by year and included in the panel one year lagged. See Table 1 for a description of all variables and their sources.

4.3.1 Dependent Variable

Our dependent variable indicates whether an economist appeared as a witness in a given hearing. *Is witness* takes the value 1 for witnesses and the value 0 for the at-risks.

4.3.2 Independent Variables

Gender. The gender of each academic witness was inferred algorithmically from their first name, using the R package gender.[3] In cases where gender could not be inferred (e.g., initials or non-US names) we assigned it manually by looking up the academic's university page, LinkedIn profile, or other digital records. We were able to identify gender for 97 percent of the sample, and the remaining are dropped from our analysis. The same approach was used for the at-risk population, where gender could be inferred for 88 percent of the sample.

Hearing format. We collect information about the hearing format from the processed hearing transcripts. Virtual hearings were typically conducted using Cisco Webex in the House of Representatives and the Senate, and this was announced in opening statements by committee Chairs.

Geography. To proxy mobility friction we estimated the travel distance and time between Washington DC and the academics' institution. The institution of each witness was inferred, using two sources: publication history and witness affiliations from hearing transcripts. The affiliations listed on a witness' stock of publications were retrieved, and a primary affiliation was chosen when multiple affiliations competed in a year. For years without affiliation information from publications, affiliations from previous years were carried forward. For the at-risk group, location was inferred from their paper affiliations. Coordinates for the affiliation were retrieved from Open Alex, and queried using the Google Maps API to retrieve the distance in miles, transit distance and transit time, and driving distance and driving time.

[3] https://cran.r-project.org/web/packages/gender/index.html

4.3.3 Control Variables

Affiliation and position. Apart from information about the academic's field, GPT-4o also identified their affiliation and position (e.g., Associate professor at MIT). As the output provided by GPT-4o contained a raw string which represents the affiliation or position of a given witness, we developed a dictionary and applied it on the original output from GPT-4o.[4] In cases for whom GPT-4o could not infer a position, we manually reviewed the economist's university page or CV to determine their position at the time of the hearing.

For the at-risk group, their institutional affiliation is inferred from their stock of publications by year. In years where the scholar publishes with more than one affiliation, the most frequent is selected. In cases of a tie, next year's affiliation is selected. In cases where next year's affiliation is missing, prioritize US based affiliations, and if all affiliations are US based, alphabetical order.

The affiliations were also used to construct the variable top ten department following the definition of top economics departments using Amir and Knauff (2008).

Academic age. Academic age is constructed at each year by subtracting the first year of publication from the year of the hearing. In OpenAlex, it occurs that scholars' profile is mistakenly accredited the work of other's, sometimes creating academic ages well above what we would realistically assume feasible. We take two approaches to mitigate the issue of these "vampires". First, we identify all authors with a gap of more than 10 years between their first and second publication and use their second publication to assign their first publication year instead of the first. Second, we drop scholars from our sample that exceed an academic age of 65, assuming that there are either disambiguation or data errors associated with these profiles.

Academic's human capital. Following previous studies of academic's external engagement (Ding et al., 2013; Tartari and Salter, 2015), we collect four measures of time variant human capital for each economist in the sample: stock of publications, stock of citations, citations/publications, average H-index, and stock of top-5 journal publications to capture prominence in the economics profession.

We proxy each academic's social connectiveness by counting unique coauthors. Coauthors are considered strong ties in academia (Azoulay et al., 2010; Ding et al., 2013). We construct a variable capturing the count of coauthors, as we assume a larger coauthor network to increase the probability of being invited to a hearing. Second, we construct a variable capturing coauthor ties to previous witnesses, as Congress members and staffers might draw on academics in their network to find other experts. We assume that direct ties to previous witnesses will be more conducive for women's probability of being selected as a witness (Ding et al., 2013).

[4] For example, from the raw affiliation string "Professor, Massachusetts Institute of Technology Sloan School of Management" we inferred that the position was "professor" and the affiliation "MIT".

Media attention and policy citations. We proxy each academic's public and policy visibility using altmetrics drawn from the stock of papers published prior to a hearing. Specifically, we measure the number of news mentions and policy citations associated with an academic's stock of publications, as these indicators capture whether their work has attracted attention beyond the academic community and entered public or policy debates. Because congressional staff and members may rely on highly visible research when identifying experts, we expect academics with greater media attention and more policy citations to be more likely to be selected as witnesses.

Hearing topic. To classify the substantive content of each hearing, we assigned JEL codes to hearing titles using a zero-shot semantic retrieval approach. For each JEL category, we constructed a semantic profile based on the official JEL classification guide by the American Economic Association,[5] combining the category name, associated keywords, and example titles into a single descriptive text. Hearing titles were embedded using a general-purpose, pretrained transformer model (all-mpnet-base-v2[6]) and compared to these profiles using cosine similarity (Song et al., 2020). We retained the top three candidate categories at each of the three levels of the JEL hierarchy: Section, sub-section, and sub-sub-section.

To ensure hierarchical consistency in the final assignment, we applied a top-down selection rule. The top-scoring section was selected first. The sub-section and sub-sub-section assignments were then chosen as the highest-scoring candidates from the top-three lists that shared the same broad category as the selected section. Where no consistent candidate existed in the top three, we fell back to the raw top-scoring prediction at that level, thus generating cases where the section-level assignment may not share a letter prefix with the sub-section (occurs in 3.8% of cases) or sub-sub-section (occurs in 14.2% of cases) assignment. This procedure relies exclusively on publicly available JEL descriptions and example titles and does not use supervised training data from the hearings themselves. Similarity scores therefore reflect semantic proximity rather than calibrated probabilities.

[5]https://www.aeaweb.org/jel/guide/jel.php
[6]https://huggingface.co/sentence-transformers/all-mpnet-base-v2

Table 1: Variable Definition and Sources of Information

Variable name	Description	Source
Is witness	Indicator equal to 1 if the academic appeared as a witness at the hearing, 0 if in the matched at-risk group	Congressional hearing records (Maher et al. (2020); Congressional Hearings (nd))
Female	1 = female, 0 = male; inferred algorithmically from first name, with manual coding for ambiguous cases	R package `gender`; manual lookup
Virtual	1 = hearing conducted in a virtual or hybrid format, 0 = in-person	Hearing transcripts
University	University affiliation at the time of the hearing, inferred from hearing transcript and bibliography	Hearing transcripts; OpenAlex
Driving distance	Driving distance in miles between the academic's university affiliation and Washington DC	OpenAlex; Google Maps API
Driving time	Driving time in hours between the academic's university affiliation and Washington DC	OpenAlex; Google Maps API
Top 10 university	1 = affiliated with a top 10 economics department, 0 = otherwise	OpenAlex; Amir and Knauff (2008) department classification
Seniority	Academic age; number of years since the scholar's first publication	OpenAlex
Citations per publication	Citation stock divided by publication stock	OpenAlex
Top 5 publication stock	Number of the academic's publications in top-5 economics journals	OpenAlex
H-Index	H-index of the academic, defined as the maximum value h such that the academic has h papers each cited at least h times.	OpenAlex
Coauthors	Count of unique coauthors in the academic's stock of publications	OpenAlex
Witness ties	Count of coauthor ties to economists who have previously testified as a witness	OpenAlex
Media mentions	Number of news mentions associated with the academic's stock of publications	Altmetric (Dimensions)
Policy mentions	Number of policy-document citations associated with the academic's stock of publications	Altmetric (Dimensions)
Hearing topic (JEL domain)	JEL code classification of the hearing topic, assigned via semantic similarity to JEL category descriptions	AEA JEL classification guide; all-mpnet-base-v2 (Song et al., 2020)

4.4 Descriptive Statistics

Table 2 depicts the gender composition of the economists in the at-risk group and the witnesses. Overall, the share of women in the at risk is larger than in the witness population.

Table 2: Number of witnesses and at-risk economists by gender

At-Risk Sample		Witness Sample	
Female	Male	Female	Male
1567 (14.5%)	9258 (85.5%)	174 (12.1%)	1268 (87.9%)

Table 3 presents descriptive statistics of the economists testifying in Congress between 1969 and 2023. Just 9 percent of witness appearances in this period were by women economists. Witnesses have tended to be senior in the field and prominent, with a mean academic age of 30 years since first publication and 32 percent being affiliated with the top 10 economics departments.

Nearly 99 percent of the sample has a U.S. university affiliation, and 10 percent are located in the

Washington DC area. The mean driving time between the witness' affiliation and Washington DC is 11 hours with a minimum of 0.3 hours and a maximum of 67 hours.

Turning to the composition of hearings in which the economists in our sample appeared, 49 percent were held by committees of the House of Representatives, 35 percent in the Senate, and 16 percent were in joint committees, in line with previous research that found hearings in the House of Representatives to be more frequent (Ban et al., 2023). These hearings are distributed unequally across committees, with committees with jurisdiction over banking and finance accounting for 21 percent, and budget and appropriations accounting for 14 percent of hearings (see appendix E for a description of our grouping of committees).

Table 3: Descriptive Statistics - Economists in Congressional Hearings

Variable	Obs	Mean	Std. Dev.	Min	Max
Female	5,293	.093	.291	0	1
Academic Age	4,684	28.428	17.102	0	99
Top 10 econ depart.	5,452	.183	.386	0	1
US affiliation	5,452	.963	.188	0	1
Washington DC Area	5,452	.103	.304	0	1
Driving time (hours)	5,284	11.194	11.73	.03	66.771
Driving distance (miles)	5,284	724.891	805.104	.23	4150.32
Chamber					
House of Representatives	5,452	.488	.5	0	1
Joint	5,452	.161	.368	0	1
Senate	5,452	.35	.477	0	1
Committee Group					
Agriculture	5,431	.067	.25	0	1
Banking and Finance	5,431	.212	.409	0	1
Budget and Appropriations	5,431	.137	.344	0	1
Economy	5,431	.184	.388	0	1
Energy and Commerce	5,431	.041	.199	0	1
Infrastructure	5,431	.034	.182	0	1
Internal	5,431	.003	.059	0	1
Judiciary	5,431	.069	.254	0	1
Labor and Welfare	5,431	.086	.28	0	1
Natural Resources	5,431	.031	.174	0	1
Other	5,431	.014	.116	0	1
Oversight	5,431	.043	.202	0	1
Science	5,431	.026	.16	0	1
Security and Foreign Affairs	5,431	.052	.222	0	1

Top 10 Departments are defined using Amir and Knauff (2008). The Washington DC area is defined as the region within 31 miles (50 kilometers) from Capitol Hill, following Deng et al. (n.d.).

Table 4 and 5 reports summary statistics of the variables separately by gender, for the at-risk and witness economists. All human capital variables are lagged one year from the time of the hearing.

Across most human capital measures, men in the at-risk group show somewhat higher averages; mean publication, citation, top-five publication stock, and average journal impact factor. When looking at the average citations per publications though, female economists are slightly higher. Male economists are more senior, on average, and have similar coauthor network counts but significantly more ties to expert witnesses – measured as co-authorship with an economist that has previously testified as an expert witness. 20 percent of female economists and 23 percent of male economists have an affiliation with a top-10 economics department at the time of the hearing, and the average driving distance to Washington is similar for the two groups.

Table 4: Summary Statistics, at-risk group

Variable	gender	N	Mean	SD	Median	Min	Max
Publication Stock	Female	4310	104.43	113.93	72.00	2.00	2468.00
	Male	39834	133.70	204.12	87.00	1.00	6370.00
Citation Stock	Female	4310	1527.11	3795.78	303.00	0.00	74111.00
	Male	39834	1862.67	5470.52	325.00	0.00	134091.00
Citations per Pub	Female	4310	9.59	15.43	4.28	0.00	390.06
	Male	39834	8.54	13.70	4.01	0.00	322.98
Avg SJR	Female	1975	2.45	2.26	1.70	0.10	12.05
	Male	10221	2.73	2.81	1.76	0.10	14.96
Top-five pub stock	Female	4310	1.70	3.81	0.00	0.00	51.00
	Male	39834	5.64	10.90	0.00	0.00	86.00
Unique Coauthors	Female	4310	58.59	86.24	27.00	0.00	597.00
	Male	39834	57.84	83.01	27.00	0.00	599.00
Witness Ties	Female	4310	1.54	3.14	0.00	0.00	33.00
	Male	39834	3.05	6.59	1.00	0.00	76.00
Seniority	Female	4310	23.96	12.93	21.00	1.00	65.00
	Male	39834	27.36	14.04	25.00	1.00	65.00
Top-ten university	Female	4310	0.20	0.40	0.00	0.00	1.00
	Male	39834	0.23	0.42	0.00	0.00	1.00
Driving distance (miles)	Female	4310	802.86	862.29	437.67	0.23	2903.16
	Male	39820	868.23	881.20	440.28	0.08	4266.46
Driving time (hours)	Female	4310	12.29	12.45	7.37	0.03	42.55
	Male	39820	13.27	12.73	7.39	0.01	69.08

Summary statistics are captured for each individual economist-hearing pair at the time of the hearing. The same economist can appear as an at-risk or witness across multiple hearings, and in that case will be included multiple times in the summary statistics.

Among witnesses, as with the at-risk group, men show somewhat higher averages on most human capital measures; publication stock, citation stock, top-five publication stock, and slightly on average journal impact factor. We observe that female witnesses have higher citations per publication. Male witnesses are also more senior on average, by around five years, and more likely to be affiliated with a top ten university (18 percent for women and 30 percent for men). Unlike in the at risk pool, where coauthor network counts are similar across genders, male witnesses have noticeably larger coauthor networks than female witnesses and also hold more ties to expert witnesses, although the difference is smaller than among the at-risk population. Geographically we observe that female witnesses are, on average, closer to Washington DC than male witnesses, the average driving time from male economists' affiliation being nearly three hours longer than women's.

Table 5: Summary Statistics, witnesses

Variable	gender	N	Mean	SD	Median	Min	Max
Publication Stock	Female	376	60.57	64.81	39.00	1.00	462.00
	Male	3626	90.66	113.79	57.00	0.00	1615.00
Citation Stock	Female	376	1248.67	2574.05	191.50	0.00	23933.00
	Male	3626	1715.80	6954.37	244.00	0.00	141118.00
Citations per Pub	Female	376	13.25	20.39	6.60	0.00	218.00
	Male	3611	9.77	17.99	4.42	0.00	349.56
Avg SJR	Female	199	2.71	2.30	2.17	0.12	12.05
	Male	889	2.91	2.85	2.12	0.10	17.20
Top-five pub stock	Female	376	1.62	2.92	0.00	0.00	20.00
	Male	3626	5.54	10.14	2.00	0.00	84.00
Unique Coauthors	Female	376	4.77	6.54	2.00	0.00	40.00
	Male	3626	7.51	10.88	4.00	0.00	95.00
Witness Ties	Female	376	3.25	5.19	1.00	0.00	35.00
	Male	3626	3.82	7.47	1.00	0.00	76.00
Seniority	Female	376	22.34	11.94	20.00	2.00	64.00
	Male	3626	27.28	14.37	25.00	0.00	65.00
Top-ten university	Female	376	0.18	0.38	0.00	0.00	1.00
	Male	3626	0.30	0.46	0.00	0.00	1.00
Driving distance (miles)	Female	375	523.29	737.70	327.06	0.32	2930.00
	Male	3596	709.85	780.20	436.96	0.23	4150.32
Driving time (hours)	Female	375	8.20	10.79	5.78	0.04	44.00
	Male	3596	10.99	11.36	7.25	0.03	66.77

Summary statistics are captured for each individual economist-hearing pair at the time of the hearing. The same economist can appear as an at-risk or witness across multiple hearings, and in that case will be included multiple times in the summary statistics.

Figure 2 depicts the number of economists and the number of hearings with witnesses by chamber between 1969 and 2023. The House of Representatives had more hearings with witnesses and a higher number of economists in their witness panels for most of the years. This is in line with their overall larger share of hearings, compared to the Senate.

It is notable but expected that the number of hearings and economist witnesses increased significantly during COVID-19. Congress passed numerous rescue plans to create and retain jobs, stimulus payments to individuals and businesses, expanded healthcare for certain groups, and more (CAR, 2020). The severe economic downturn, including increases in unemployment and business closures, is likely to simultaneously have increased the need for more hearings and the use of economists' expertise to navigate complex policy development and ensure evidence-based decision making.

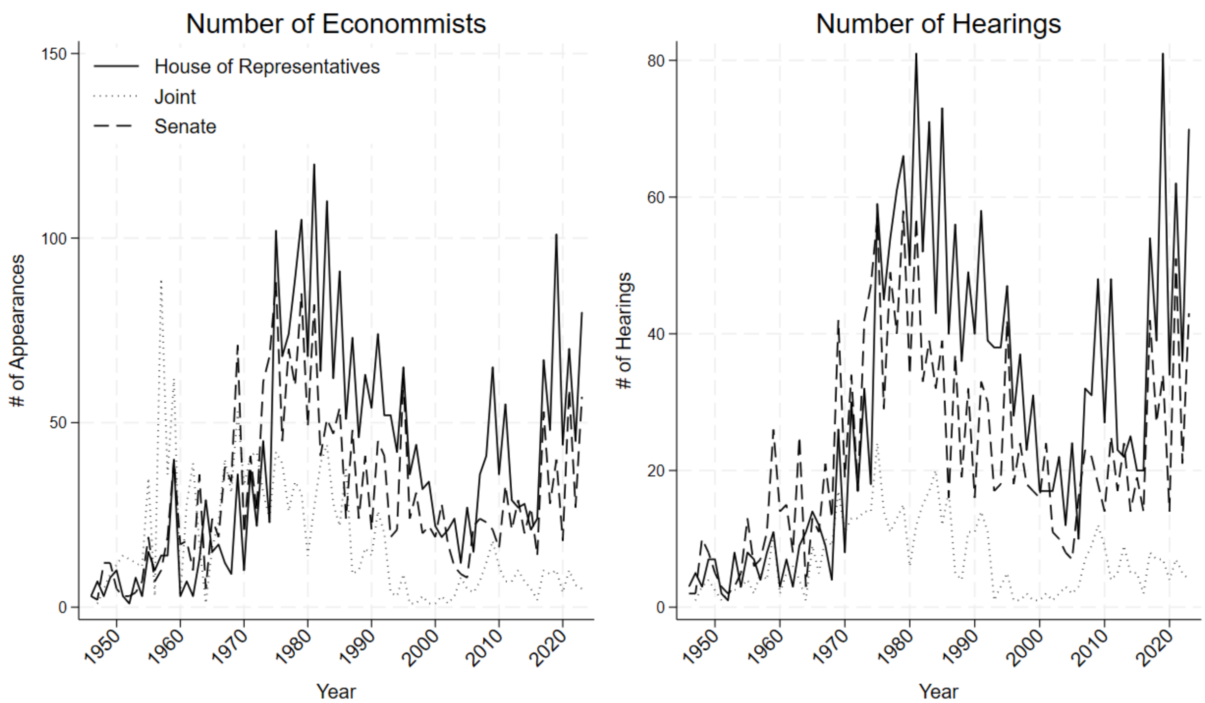


Figure 2: Number of Economists and hearings with economists as witnesses, by Chamber

Turning to the geographical diversity of witnesses, Figure 3 depicts a map of witness proportion by state from 1969-2023, based on the university affiliation. Two facts seem apparent from the map; that researchers from the east-coast are heavily represented, and that there is a concentration around the location of top economics departments. Apart from being home to a number of elite universities, the East-Coast is better connected to Washington D.C. by transportation (e.g. train connectivity, as well as shorter travel times due to geographic proximity), making it easier for researchers to travel to and from Congress. The apparent strong representation of elite institutions is in line with literature suggesting that the field of economics is characterized by a somewhat homogenous elite that concentrates around its' top departments, compared to other fields (Korom, 2020). Finally, it is noteworthy that aside from California in the West Coast, and Illinois in the Midwest, economists from most US states, including large and populated ones such as Texas and Florida, are historically less represented in Congressional hearings. To expand on these findings, we are planning to explore historical trends in representation of different institutions. These findings add to the literature on institutional concentration of power and hierarchy in the economics profession (Fourcade et al., 2015; Freeman et al., 2024; Hirschman and Berman, 2014; Jones and Sloan, 2021), with insights from a previously underexplored and highly impactful aspect of policymaking engagement.

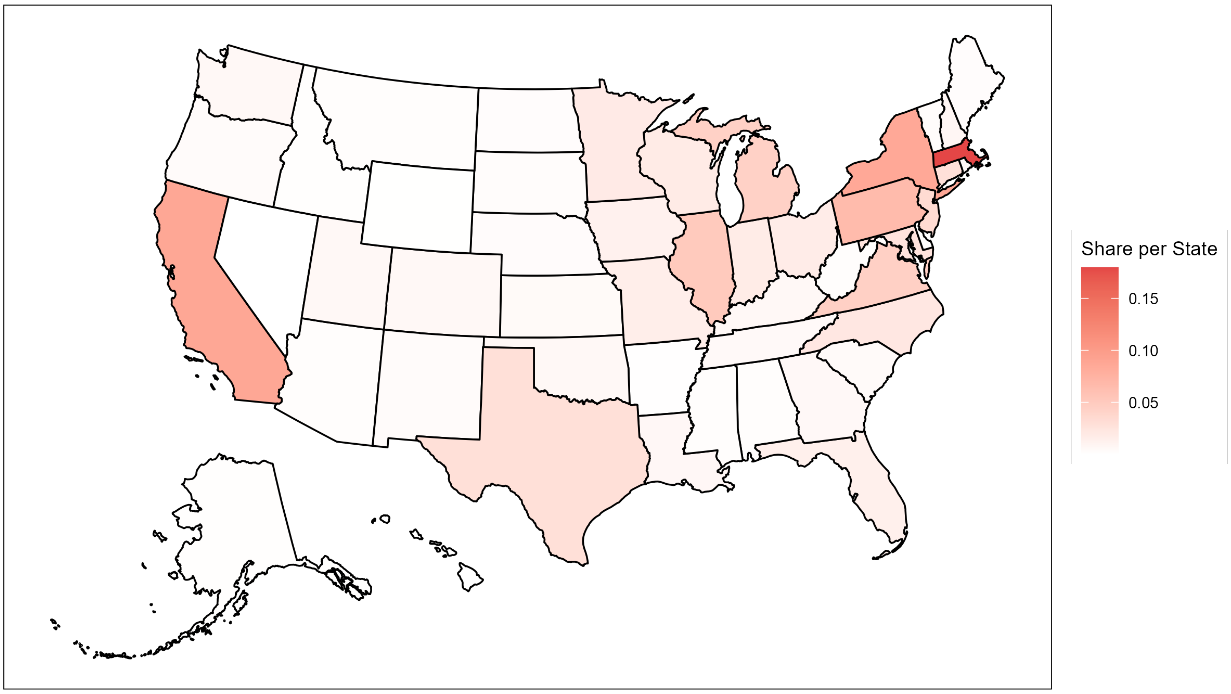


Figure 3: Share of economists serving as expert witnesses in congressional hearings by state (1969-2023)

Note: The data from 1969-2016 is from Maher et al. (2020) and the data from 2017-2023 is from our data collection, detailed in section 4.1.1 and 4.1.2.

Turning to the content of hearings, figure 4 presents the distribution of testimonies across broad JEL topic areas. Women make up 10% of witness appearances over time but are represented unevenly across different subject areas. We observe the largest representation of women, 20%, in hearings on issues related to micro- and labour-economics.

This pattern is consistent with a well-documented gender segregation in economics research fields, where women are disproportionately represented in health, education, and labour economics relative to macroeconomics and finance (Antman et al., 2024; Beneito et al., 2019). This topic-level gender gap in congressional testimony could reflect supply-side sorting, whereby women are less likely to be active researchers in macro and finance and therefore less likely to be invited, or demand-side factors, whereby committees seeking gender diversity preferentially recruit women for hearings on topics where they are relatively more represented.

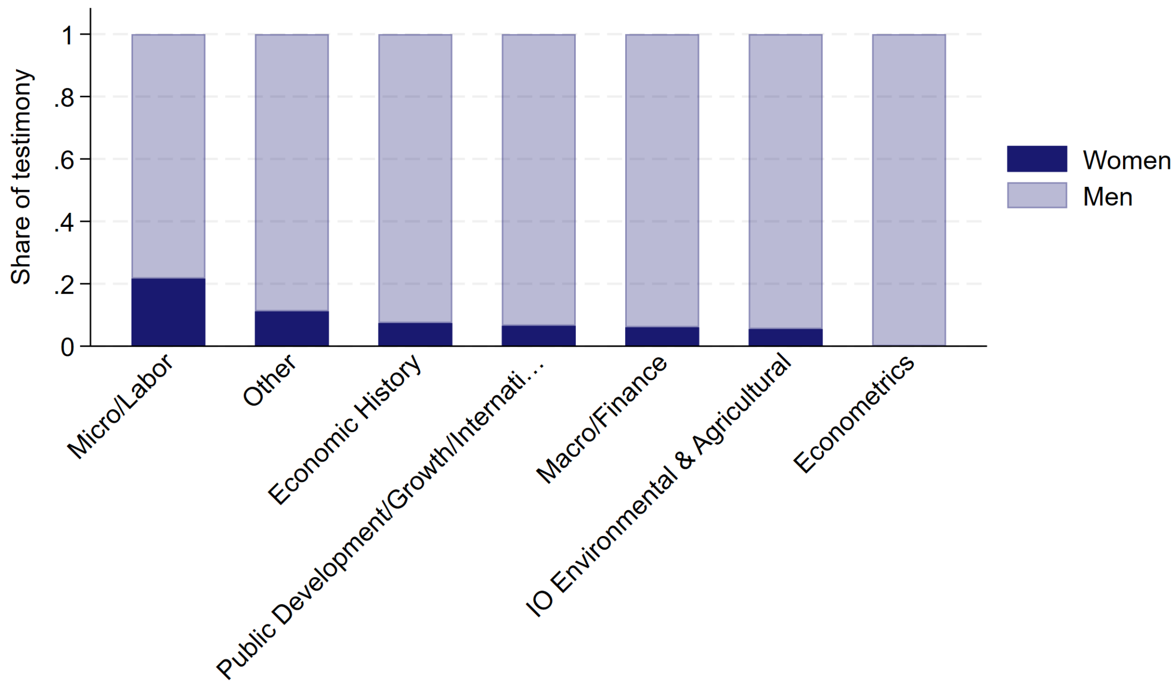


Figure 4: Share of women witnesses across JEL sections.
JEL codes are grouped following Singhal and Sierminska (2024)

Regarding the number of hearings per economist, certain individuals are accounting for a large share of witness appearances. Appendix B presents the most frequently appearing economists between 1969 and 2023.[7] The list features prominent economists, including four Nobel Prize winners, and seven members, presidents, or chairmen to the President's Council of Economic Advisors.[8] There is one woman on the list.[9]

Finally, Appendix C is a list of the most common university affiliations, both in terms of the number of individual witnesses and the total number of testimonies. When looking at the most common university affiliations by number of testimonies, Harvard University accounts for nearly 10 percent of the economists testifying, followed by MIT and University of Pennsylvania, which accounts for 4.5 and 3.3 percent, respectively. The list features prominent universities, which is unsurprising, considering the aforementioned concentrated nature of the profession (Ductor and Visser, 2023; Freeman et al., 2024; Korom, 2020, e.g.,). Some rankings on the list are, however, surprising.

[7]Simon Johnson (MIT), Lawrence Klein (MIT), James Tobin (Harvard University), Paul Samuelson (Harvard University)

[8]Martin Feldstein (Chair), Walter Heller (Chair), Murray Weidenbaum (Chair), John Taylor (member), Hendrik Houthakker (member), Alan Blinder (member), James Tobin (member)

[9]Karen Davis (The Johns Hopkins School of Public Health)

The University of Maryland is ranked before the University of Chicago, both in terms of absolute hearings and unique witnesses. This might suggest that proximity is more important than prestige. However, Duke University is ranked low on the list, despite that it is high ranking amongst top economics departments and its proximity to Congress.

When considering different universities' representation in Congress hearings, there could be concerns that this is driven by a small group of active economists in that institution. Appendix C also offers a list of the share of unique economists per university. The same universities largely dominate this picture as well, but with some differences in their share. For instance, MIT accounts for 4.5 percent of testimonies but only 1.82 percent of unique witnesses.

5 Methods

5.1 Temporal trends

Our first approach examines temporal patterns in congressional hearing participation through a linear probability model (LPM). While a nonlinear model might seem more appropriate given our binary outcome variables, we opt for an LPM due to its straightforward interpretation of coefficients and its reliability in estimating marginal effects, particularly with multiple interaction terms (Wooldridge, 2010). We estimate:

$$Y_i = \beta_0 + \beta_1 \lambda_{it} + \beta_2 Chamber_i + \beta_3 Chamber_i \times \lambda_t + \varepsilon_i \tag{1}$$

where Y_i represents various witness characteristics for appearance i. We examine three key outcome variables: whether the witness is female, whether they are affiliated with a non-elite institution, and whether they are affiliated with a remote institution. The model includes year fixed effects λ_t to capture aggregate temporal trends and chamber indicators $Chamber_i$ to account for systematic differences between House, Senate, and Joint committee hearings.

In some specifications, particularly when examining subgroups with limited observations, we employ Congress-level rather than year fixed effects to ensure sufficient statistical power. This approach aggregates observations into two-year periods, trading some temporal precision for increased stability in our estimates. The interaction term $Chamber_i \times \lambda_t$ allows the temporal patterns to vary by chamber. ε_i denotes the error term. We employ robust standard errors to account for heteroskedasticity.

To visualize these patterns, we compute and plot the marginal effects of the predicted probabilities for each year. These plots provide an intuitive representation of how participation patterns evolved over time, while controlling for direct and chamber-specific effects.

5.2 Main analysis

Our main analysis estimates a conditional logit model to examine the probability that an economist is selected to testify before Congress relative to a matched pool of plausible alternatives. For every hearing, we construct a stratum consisting of the realized witness or witnesses and the matched at-risk economists identified through the semantic similarity procedure described in section 4.2. The unit of analysis is therefore the academic-hearing pair, where the dependent variable, Y_{ih}, equals one if academic i was selected as a witness for hearing h, and zero otherwise.

Formally, the conditional probability that economist i is selected as a witness in hearing h is given by:

$$P(Y_{ih} = 1 \mid h) = \frac{\exp(X_{ih}\beta + \eta_h)}{1 + \exp(X_{ih}\beta + \eta_h)} \tag{2}$$

where h indexes the hearing-specific choice set and X_{ih} denotes economist-level characteristics at the time of the hearing and η_h denotes the hearing strata. Because the model conditions on the hearing stratum, all hearing-level characteristics are absorbed by the conditional likelihood. Identification therefore comes entirely from within-hearing variation between the selected witness and the matched at-risk economists associated with the same hearing. This design controls for both observed and unobserved hearing-specific factors, including committee composition, hearing topic, political salience, and the overall demand for expert testimony.

The fully interacted estimating equation is specified as:

$$\begin{aligned} X_{ih}\beta = {} & \beta_1 Female_i + \beta_2 Virtual_h + \beta_3 \log(DrivingTime_{ih} + 1) \\ & + \beta_4 Female_i \times Virtual_h + \beta_5 Female_i \times \log(DrivingTime_{ih} + 1) \\ & + \beta_6 Virtual_h \times \log(DrivingTime_{ih} + 1) + \beta_7 Female_i \times Virtual_h \times \log(DrivingTime_{ih} + 1) \\ & + \gamma Z_{ih} \end{aligned} \tag{3}$$

where $Female_i$ is an indicator equal to one for women economists, $Virtual_h$ indicates hearings conducted during the COVID-19 period in which hearings were held virtually, and $\log(DrivingTime_{ih})$ measures the logged driving time from the economist's institutional affiliation to Washington DC. The vector Z_{ih} includes controls for citations per publication, cumulative coauthors, seniority, top 10 university affiliation, and stock of top five publications. All controls are measured in the year prior to the hearing.

The interaction between female and driving time captures whether geographic distance imposes a larger penalty on women economists relative to men in the selection process. The triple interaction between female, virtual hearings, and driving time examines whether the transition to virtual hearings reduced this differential distance penalty for women. A positive coefficient on the triple interaction would indicate that virtual hearings disproportionately increased the likelihood that geographically distant women economists were selected to testify, consistent with the interpretation that mobility constraints constitute an important barrier to women's participation in congressional testimony.

Because coefficients from conditional logit models are expressed in log-odds and are not directly interpretable in probability terms, we compute average marginal effects and predicted probabilities to aid interpretation of the substantive magnitude of the estimates, particularly for interaction terms. Marginal effects are calculated from the fitted conditional logit model while holding all other covariates at their observed values.

6 Results

6.1 Temporal Patterns in Congressional Testimonies

The following section provides descriptive evidence of time trends of the representation of female economists in hearings. This analysis uses only the sample of economists who have testified in hearings, not the full panel dataset.

Figure 5 depicts plotted marginal predictions from a regression of the variable female on an interaction between year and Congress chamber. The plot shows low probability of women to appear in hearings before 1980. This is unsurprising, given that women have historically been underrepresented in the economics profession, especially in higher positions. After 1990, there seems to be an upward trend in the probability of women economists to give testimony, with a particularly steep increase in the probability after 2019, when the in-person hearings changed to an online format.

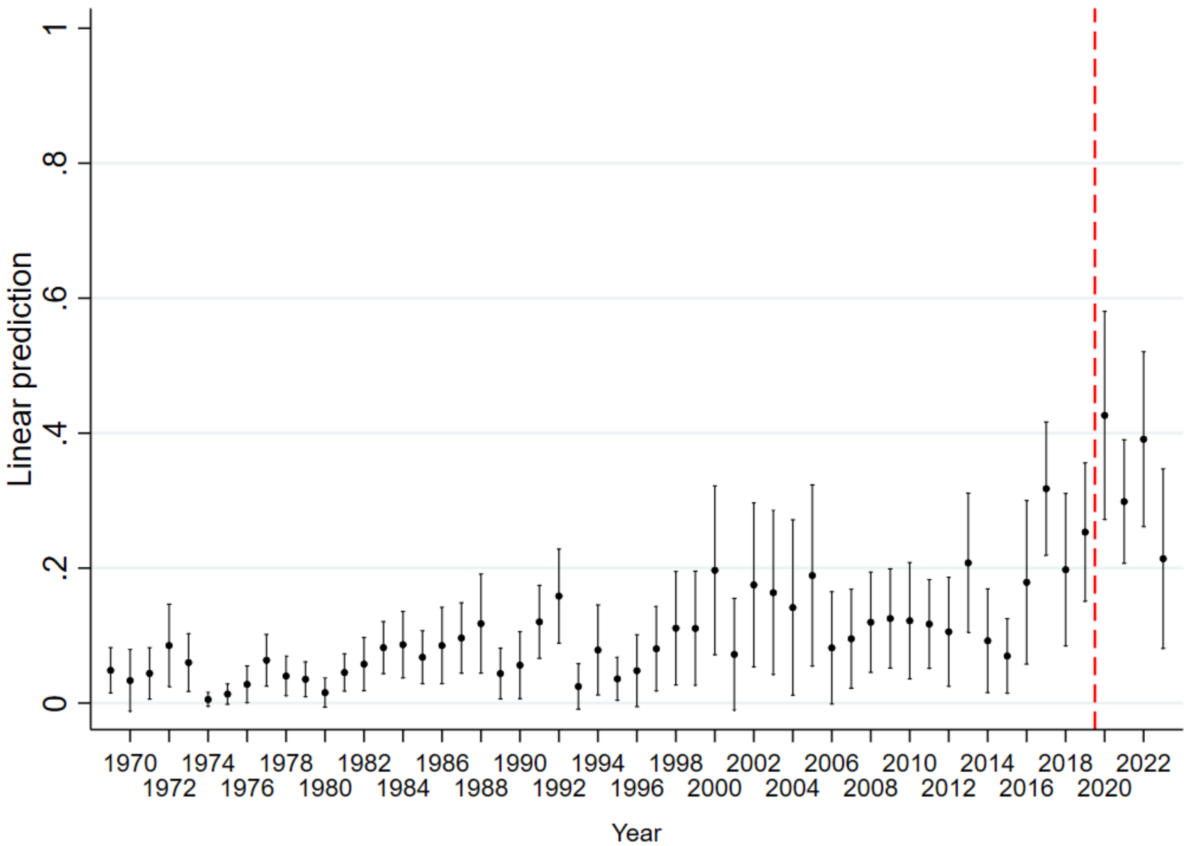


Figure 5: Female Economists' Participation in Congressional Hearings

Marginal predictions from year and Congress chamber interaction regressed on female with robust standard errors. Dashed line indicates onset of online hearings.

Figure 6 depicts marginal predictions of the probability of economists from elite universities to participate in hearings, compared to those in non-elite universities. The coefficients show some variation over the years, but there seems to be no clear time trend with the confidence intervals overlapping most of the years. There also seems to be no significant increase or decrease after the change to remote hearings in 2020. As discussed in Section 4.4, elite institutions are overrepresented in our sample.

However, when examining compositional changes within non-elite institutions, a different pattern emerges. While the overall participation of economists from non-elite institutions to testify in Congress did not increase following the shift to virtual hearings, the composition of who testifies from these institutions changed substantially. Specifically, female economists from non-elite departments saw a marked increase in their hearings' participation around 2016, with this trend accelerating significantly after the transition to virtual hearings in 2020. This suggests that while virtual hearings did not break down the barrier between elite and non-elite institutions, they may have reduced barriers to participation for women based in non-elite institutions.

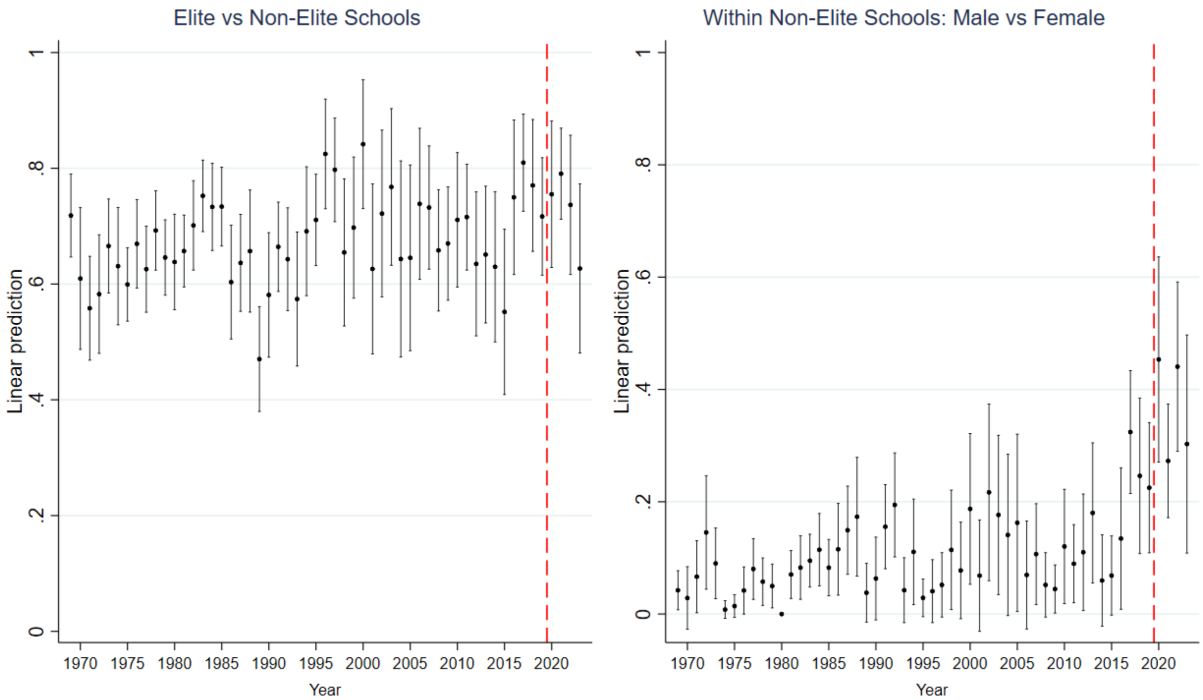


Figure 6: Participation in Hearings of Economists from Elite vs. Non-Elite Universities
Year and Congress chamber interaction regressed on non-elite department affiliation (non-elite = 1), with robust standard errors.

Following, we move to examine trends in the participation of economists based in remote schools, displayed in Figure 7. The left side of Figure 7 shows the participation in congressional hearings of remote economists by year. There seems to be a slight downward trend which breaks around 2014, with no visible break around the shift to virtual hearings. Similar to Figure 6, the right side of figure 7 examine whether a compositional change took place (between female economists compared to male) within the sample of remote economists. There seems to be slight upward trend in remote female economists' participation in hearings, which increases more sharply just before and after the COVID-related policy change. It seems that following the shift to virtual, the participation of women in remote institutions increased relative to men.

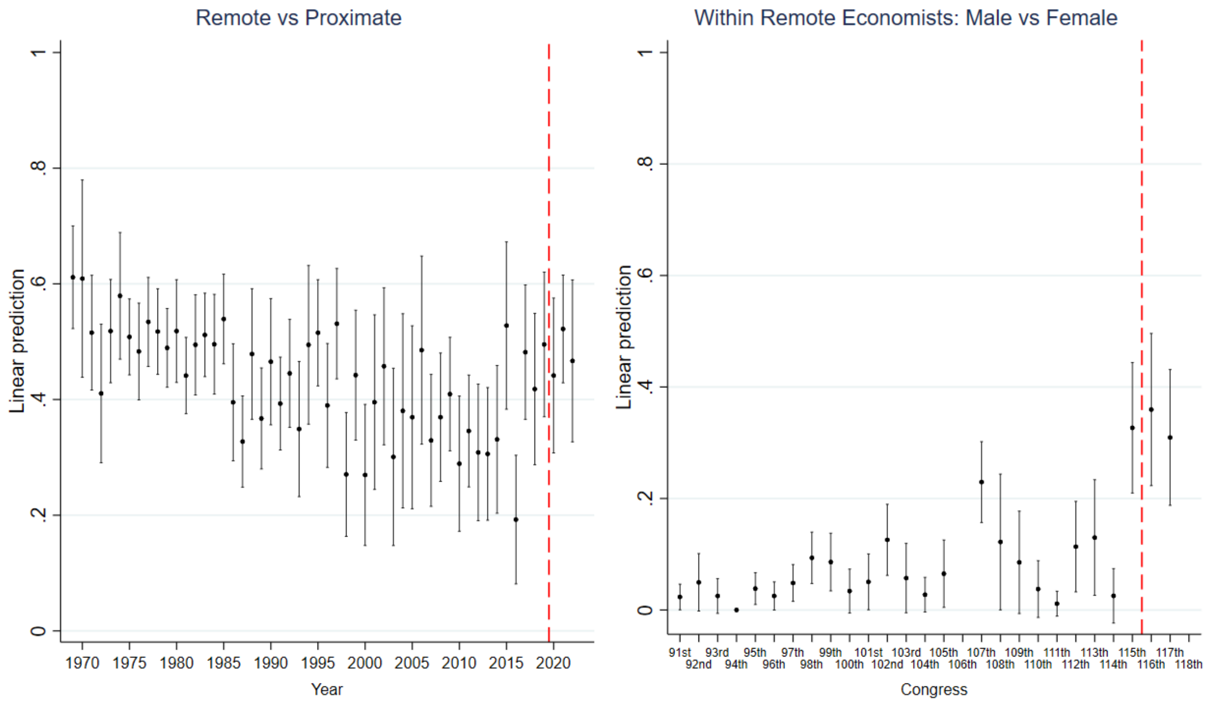


Figure 7: Participation in Hearings of Remote vs. Proximate Economists

Virtual hearing and committee group interaction regressed on proximity to Congress (remote = 1), with robust standard errors. Analysis is conducted at the year level for the full sample (left panel), while gender comparisons within remote institutions (right panel) are aggregated by Congress (two-year periods) to address small sample concerns.

We now explore gender differences in congressional witness testimony activity, by different committee types and between in-person and virtual hearings (see appendix E for classification of categories). As can be seen from the left side of figure 8, the marginal effects by different committees mostly cluster together around zero, with Labor and Welfare hearings being more positively associated with being female. This could be a result of the documented higher proportion of women in microeconomics fields, compared to macroeconomics and finance (Chari and Goldsmith-Pinkham, 2017). The marginal effects for virtual hearings show more variation, with Labor and Welfare, Internal, and Judiciary committees being more positively associated with women participating in hearings. However, as shown in Table 3 the shares of hearings in these three categories are quite low, which warrants caution when interpreting these findings.

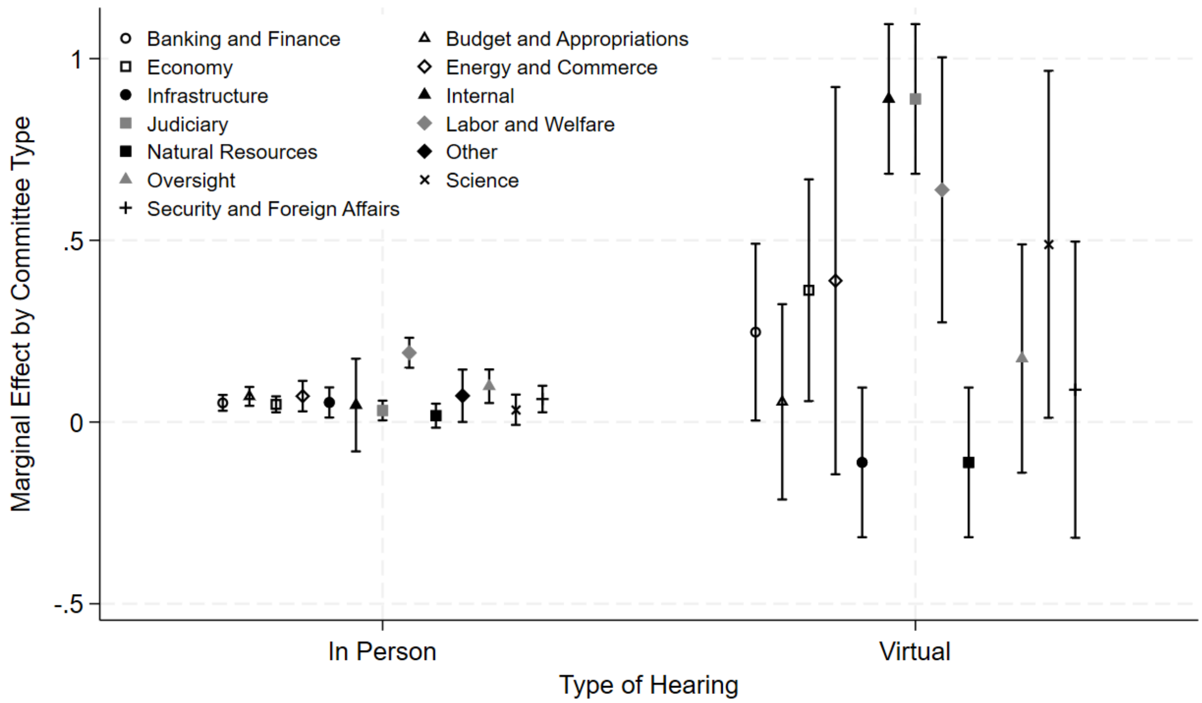


Figure 8: Marginal Effects of Virtual Hearings on Participation of female Economists, by Committee Type

Virtual hearing and committee group interaction regressed on gender (female = 1), with robust standard errors.

In conclusion, analyzing the congressional hearings dataset reveals time trends in female economists' participation in congressional hearings. We document an upward trend in participation since 1969, in line with rising levels of women in the profession. After the COVID-19 induced change to hearing formats from in-person to online, the women sharply increased their participation in hearings. We show that the shift to virtual hearings did alter the time trends in the participation of academics from non-elite institutions, and remote ones. However, this overall trend masks the increased participation of women based these institutions. We also showed that the shift to virtual hearings particularly increased the number of women giving testimony online in committees with jurisdiction over Labor and Welfare, Judiciary, and Internal areas.

6.2 Main results

Table 6 reports the results from the conditional logit models where the outcome is an indicator for being the chosen witness, with hearings serving as the choice set. This design compares each actual witness against the pool of eligible at-risk economists for that specific hearing, controlling for all hearing-level characteristics by construction, and taking into account the potential pool of matched experts per hearing. Coefficients are reported as log-odds.

Model 1 estimates the effect of gender on the probability of appearing as a witness without controls. The coefficient is close to zero and insignificant. This suggests that conditional on being in the choice set for a given hearing, women are not significantly less likely to be selected.

Model 2 adds controls for researcher characteristics. The female coefficient becomes negative and significant, indicating that conditional on observable characteristics such as citation impact, seniority, co-author network size, and top-five publication stock, women have approximately 13 percent lower odds of appearing as a witness.

Model 3 introduces log-transformed driving time between Washington DC and the economists' affiliation, which enters negative and highly significant, indicating that conditional on being in the eligible pool for a hearing, economists located further from Washington DC are significantly less likely to be selected as witnesses. Specifically, a one unit increase in log driving time is associated with approximately 16 percent lower odds of appearing as a witness. Controlling for driving distance increases the female coefficient in size and significance, indicating that the gender gap is not driven by geographic sorting of women and men relative to Washington DC.

Model 4 introduces the interaction between the gender of the economist and the hearing format. The interaction coefficient is large, positive, and highly significant, suggesting that the shift to virtual hearings dramatically increased the probability of a woman appearing as a witness relative to men, to a level that more than offsets the gender gap in the in-person hearings.[10] The coefficients are plotted in figure 9.

[10] Note that the main effect of virtual hearings drops out, as this is constant within hearings, and therefore absorbed by the hearing strata.

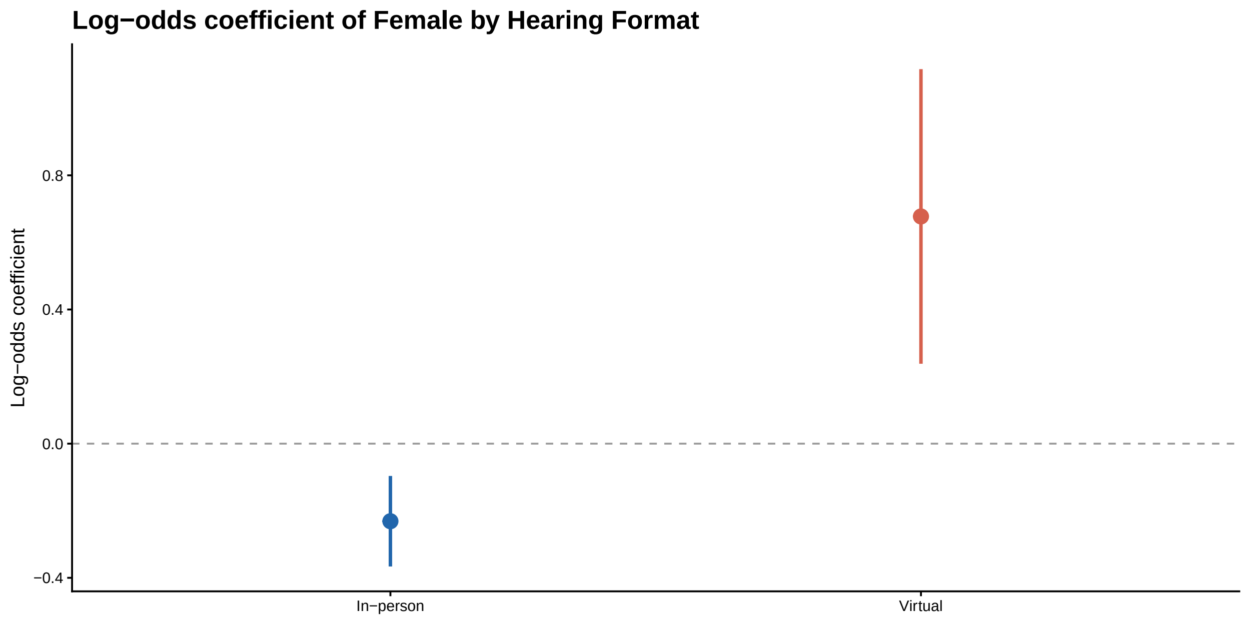


Figure 9: Gender Gap by Hearing Format

The gender gap in the odds of appearing as expert witness before Congress, by hearing format. The plotted coefficients are from an interaction between female x hearing format, with all control variables.

Model 5 examines whether the female disadvantage in selection is larger among geographically remote scholars. The interaction between female and log driving time is negative and highly significant suggesting that women face more friction related to larger geographical distances, compared to male economists. The main effect of the female coefficient, which captures the effect for women closest to Washington, is now positive and significant, suggesting that women closest to Washington have an advantage over male economists in appearing as a witness. Figure 10 plots the predicted odds of testifying for women relative to men, at different distances to Washington DC. The odds decrease for women as distances increases.

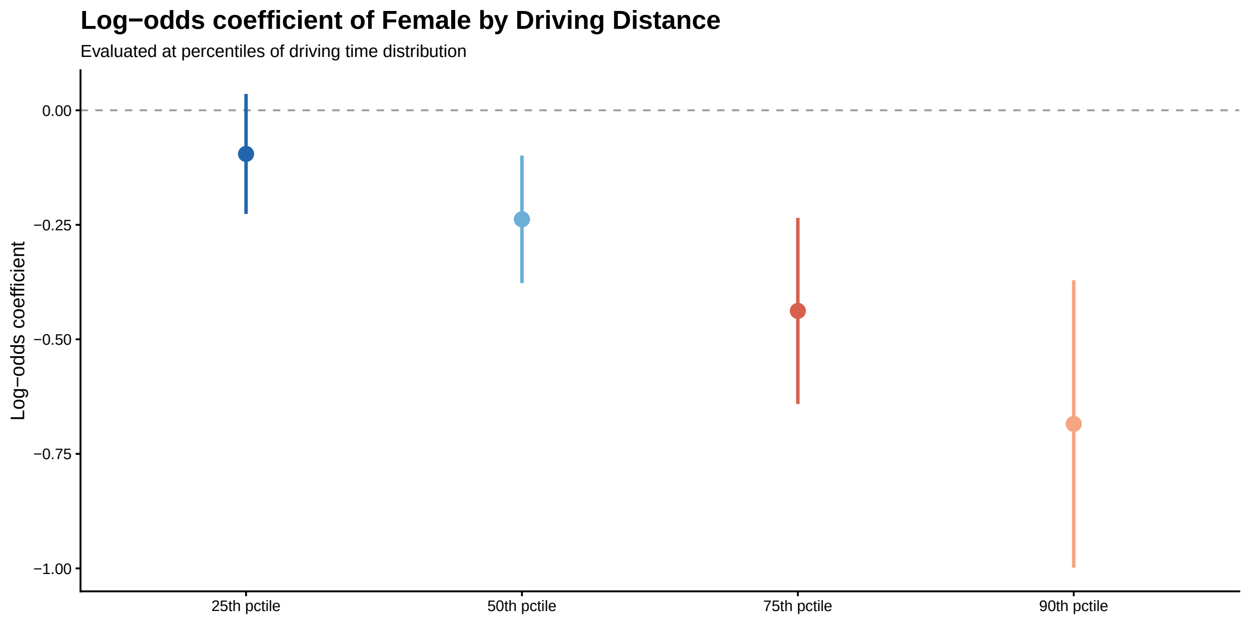


Figure 10: Gender Gap by Distance to Washington DC

The gender gap in likelihood to testify by driving distance between academic's affiliation and Washington DC. The coefficients are derived from a conditional logit with an interaction between female × driving time, with estimated coefficients at the 25th, 50th, 75th, and 90th percentile. The graph shows the increase in the gap as distance increases.

Column 6 includes a female × driving time × hearing format interaction. The triple interaction coefficient is positive and significant, consistent with the interpretation that virtual hearings disproportionately benefited women located further from Washington DC. Figure 11 plots this relationship, illustrating that women are less likely to appear as witnesses in the in-person hearings when they are furthest from Washington, and even more likely to appear in the virtual formats. The advantage for women further from Washington to appear in the virtual formats could be a substitution effect; the virtual hearings unlocked previously inaccessible talent in remote institutions, and policymakers substituted these with experts around Washington.

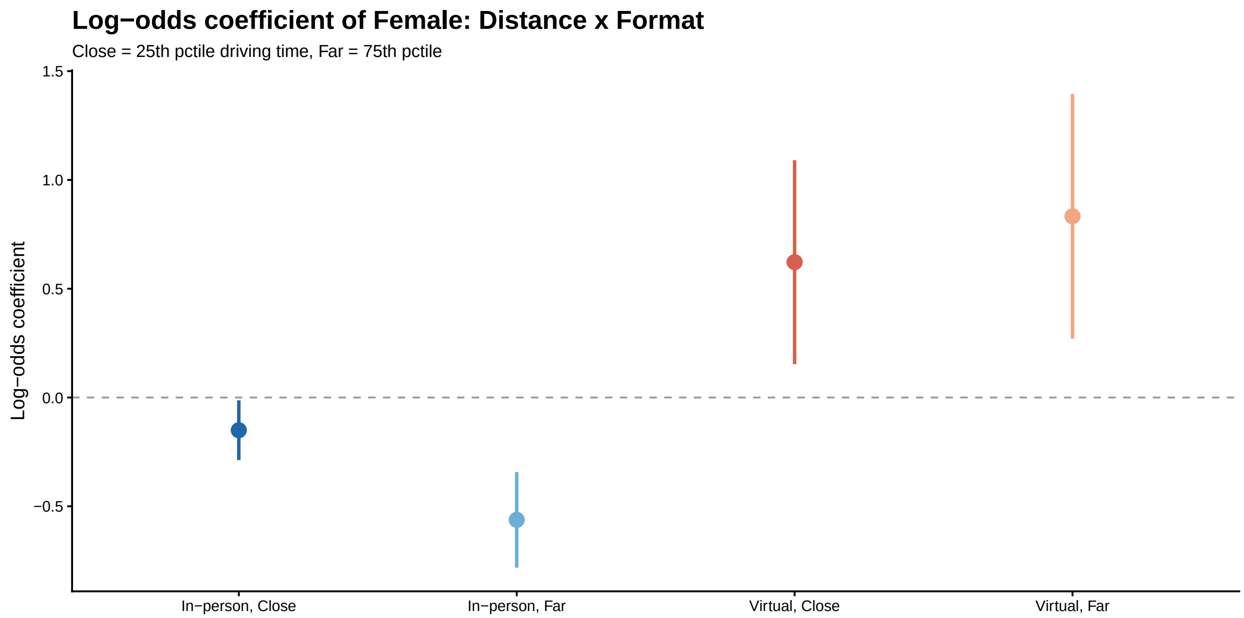


Figure 11: Gender Gap by Distance to Washington DC and Hearing Format

The gender gap in likelihood to testify by driving distance between academic's affiliation and Washington DC and hearing format. The coefficients are derived from a conditional logit with an interaction between female × driving time × hearing format, with estimated coefficients at the 25th and 75th percentile driving distance from Washington, and by hearing format. The gap is largest for economists located further from Washington.

Table 6: Conditional Logit Stratified on Hearing

	(1)	(2)	(3)	(4)	(5)	(6)
Female	-0.01	-0.13^{*}	-0.17^{**}	-0.23^{***}	0.37^{*}	0.41^{**}
	(0.06)	(0.07)	(0.07)	(0.07)	(0.15)	(0.16)
Log Driving Time			-0.18^{***}	-0.18^{***}	-0.15^{***}	-0.15^{***}
			(0.02)	(0.02)	(0.02)	(0.02)
Virtual						
Female x Virtual				0.91^{***}		-0.07
				(0.23)		(0.52)
Female x Log Driving Time					-0.29^{***}	-0.34^{***}
					(0.08)	(0.08)
Virtual x Log Driving Time						-0.21
						(0.14)
Female x Virtual x Log Driving Time						0.52^{*}
						(0.23)
N	48062	48062	48062	48062	48062	48062
Events	3967	3967	3967	3967	3967	3967
Clusters	2955	2955	2955	2955	2955	2955
Log-likelihood	-11354.73	-7645.49	-7613.51	-7607.60	-7606.61	-7597.96

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$; $^{\cdot}p < 0.1$

Coefficients are reported as log-odds. Standard errors are in parentheses. The conditional logit model stratifies by hearing, comparing each actual witness against the pool of eligible at-risk economists for that hearing. Full table with control coefficients can be found in appendix F.

7 Discussion and Conclusion

This paper documents the gender gap in academics' participation in federal level policy advising and explores how mobility friction drives this gap. Previous work has focused mainly on gaps in STEM scientists' diffusion of knowledge to industry and documented mechanisms on the supply- and demand-side. Our paper shifts the focus to another crucial channel of knowledge diffusion, from academia to government, and explores a largely overlooked constraint; that expert engagement is a spatially concentrated activity, and women face systematically higher costs of geographic mobility.

Extant studies have documented the importance of geographic proximity in knowledge transfer and collaboration (Jaffe et al., 1992; Katz, 1994) and finds that reduced geographic friction increases collaboration (Catalini et al., 2020). We also know that women tend to experience more geographic friction than men, evident in sorting into jobs with better work-life balance (Barbulescu and Bidwell, 2013), temporal flexibility (Goldin, 2014), and shorter commutes (Barbanchon et al., 2021). Our study contributes to this discussion by documenting that mobility friction not only contribute to the job segregation that drives a large proportion of the gender wage gap but also impedes women's participation in high-stakes professional opportunities, once they have sorted into a job.

We complement (Ding et al., 2010) who finds that the introduction of information technologies between 1969-1993 benefit women scientists' collaboration and productivity patterns more than men, and Catalini et al. (2020) who show that simply reducing travel costs through the introduction of cheaper airfares does not benefit women to a greater extent than men. Our findings establish that what matters to gender disparities in participation is indeed not the cost of travel but whether travel is required at all. Virtual formats that eliminate the need to be physically present produce large gains in women's participation, consistent with the intuition that women are less mobile due to family and care obligations. We expand this from scientific collaborations to a setting of high-stakes, episodic, expert advising to policymakers.

Our analyses show that women academics have lower odds of appearing as an expert witness before Congress compared to men. With controls for the academics' human capital, social network, geographic distance to Washington DC, and elite institutional affiliation, we find that women have around 13 percent lower odds of appearing as a witness. We complement prior art that finds significant and persistent gender gaps in academics' engagement with industry, notably patenting (Whittington and Smith-Doerr, 2008), university-industry collaborations (Tartari and Salter, 2015), academic entrepreneurship (Abreu and Grinevich, 2017), and science advisory boards (SABs) (Ding et al., 2013). Ex ante, it is not clear that we would find a similar gender gap in women's engagement with policy advising. The crucial role of social networks in generating commercial opportunities has been well documented (Murray and Graham, 2007), especially in opportunities that arise by invitation (Ding et al., 2013). In contrast to industry sectors relevant to commercialization opportunities, women are overrepresented in the public sector, potentially advantaging the invitation of women (OEC, 2025). Our findings establish that the underrepresentation of women in high-impact academic engagement is not confined to industry-facing activities but extends to the channels through which scientific expertise reaches federal policymakers. This is notable because the stakes and visibility of policy engagement differ from commercialization. Rather than creating private value, expert testimony shapes the information environment of federal legislators on matters of public consequence.

To assess the role of mobility constraints on the gender gap, we leverage an exogenous change in the hearing format in Congress during 2020-2023. Our results show that the virtual hearing formats not only reduced the gender gap but offset it to the point where women became more likely to testify. As the conditional logit takes out hearing characteristics such as topics and committee composition, we can isolate the effect of the hearing format. These effects were stronger for women further from Washington, for whom the odds of appearing as a witness in the in-person hearings were even smaller, and the odds of appearing online even bigger.

Our study contributes to three main streams of literature. First, we contribute to the literature on geography and knowledge work, which has shown that geographic frictions shape collaboration, knowledge transfer, and innovation (e.g., Jaffe et al., 1992; Katz, 1994; Ding et al., 2010; Catalini et al., 2020). Existing evidence, however, paints an incomplete picture. While reducing communication costs disproportionately benefits women scientists' collaborations and productivity (Ding et al., 2010), reducing travel costs appears to benefit primarily men (Catalini et al., 2020). We help explain these findings by studying expert advisory work, a setting in which communication can occur remotely but participation has traditionally required physical presence. Our findings suggest that an important dimension of geographic friction is not the cost of travel alone but the organizational requirement for in-person participation, highlighting how the design of knowledge work shapes who can participate in it.

Second, we contribute to the academic engagement literature by advancing understanding of expert advisory work as an important yet relatively understudied form of engagement between academics and organizations (Perkmann et al., 2013, 2021). We document substantial gender disparities in participation in these influential advisory roles and identify geographic mobility frictions as an important structural mechanism underlying these disparities. Finally, we show that communication technologies that reduce the need for physical presence can mitigate these frictions and increase the participation of women in expert advisory work. Together, these findings extend existing accounts of academic engagement by highlighting how the organization of advisory work shapes who participates in it.

Third, we contribute to the literature on geographic mobility and gender inequality in labor markets, which has shown that differences in commuting, relocation, and workplace flexibility shape occupational sorting, career progression, and earnings (Blackburn, 2010; Goldin, 2014; Mincer, 1978; Peterson, 1979; Sandell, 1977; Schoenbaum, 2017; Sorenson and Dahl, 2016; Widiss, 2012; Borghorst et al., 2024; Casado-Díaz et al., 2023; Barbanchon et al., 2021; He et al., 2021; Wiswall and Zafar, 2018). We extend this literature to episodic professional opportunities that require temporary travel rather than permanent or long-term relocation. Our findings show that geographic mobility frictions influence access to prestigious professional opportunities even when mobility demands are infrequent and short-lived, suggesting that the consequences of mobility constraints extend beyond employment choices to the allocation of influential advisory roles.

8 Limitations and Future Work

Our study has several limitations. First, congressional testimony is a relatively rare event and only one of many channels through which academics engage with policymakers. While it offers unusual advantages for causal identification, generalizing our findings to other forms of expert advisory work, consulting, agency panels, or informal briefings, remains an open question. Future research should examine a broader range of engagement channels and a wider set of scientific fields to assess how far these dynamics extend beyond congressional hearings and economics.

Second, our at-risk population is constructed using semantic similarity between publications, which approximates, but cannot perfectly recover, the true set of experts committees actually considered. Although this approach is validated against observed witness characteristics, some plausible candidates may be omitted and some included candidates may not have been genuinely considered, which could attenuate or bias our estimates in ways we cannot fully sign.

Third, our design does not fully disentangle preference from opportunity. The shift to virtual hearings helps identify an exogenous reduction in mobility costs, but we cannot observe whether the resulting change in women's participation reflects greater willingness to accept invitations, a greater likelihood of being invited, or both. Data on invitations rather than only realized testimony would allow future work to separate these margins directly.

Appendix A: Prompt and Example Provided to OpenAI for the Extraction of Witnesses

```
example_json = {
    "hearings": [
        {"year": "2017",
         "date_begin": "2017-02-28",
         "date_end": "2017-07-18",
         "virtual_hearing": "no",
         "full_name_string": "Williams, Ph.D., Gary W.",
         "witness_name": "Gary W. Williams",
         "witness_affiliation": "Professor of Agricultural Economics and Co-Director, Agribusiness,
         Food, and Consumer Economics Research Center, Department of Agricultural Economics, Texas A
         &M University",
         "job_title": "Professor, Co-Director",
         "witness_title": "Ph.D.",
         "academic_witness": "Yes",
         "discipline": "Economics",
         "university_affiliation": "Texas A&M University",
         "full_committee": "Committee on Agriculture",
         "sub_committee": "Subcommittee on Livestock and Foreign Agriculture",
         "sub_committee_count": "6",
         "title_description": "The Next Farm Bill",
         "chamber": "House of Representatives"
        }
    ]
}

def get_messages(testo):
    return [
        {"role": "system", "content": "You are tasked with extracting all witness information from
        the following congressional hearing text. Output the witness details as JSON, with each
```

```
        witness being represented by a dictionary. Be thorough: witnesses might be listed
        consecutively or their details may be spread across lines. Typically they will appear in a
        consecutive list after ’WITNESSES’ and be separated by a list of dots ’.......’. When you
        think you are done, check again to ensure no witness is missed. Use this schema for each
        witness: "+json.dumps(example_json)},

{"role": "user", "content": f"Extract the following information from the provided congressional
hearing text. Each witness should be represented in this JSON structure: [{{’year’:’the year of the
 hearing’,’date_begin’:’the first day of the hearing’,’date_end’:’the last day of the hearing’,’
virtual_hearing’:’yes when the hearing was a hybrid or online format and otherwise no.’,’
full_name_string’:’the full name of the witness. Make sure you exclude the affiliation here.’,’
witness_name’:’full name of the witness in the format first name middle name last name’,’
witness_affiliation’:’The full affiliation of the witness. If there is more than one affiliation,
list them all.’,’job_title’:’The job title of the witness, inferred from their affiliation. If
there are more, list them all separated by a comma.’,’witness_title’:’if there is a title in the
name, such as honorable, PhD, Dr, or Jr., save it here. If there are more than one title, list them
 all.’,’academic_witness’:’yes if the witness works as an academic, otherwise no. The mention of
professor and/or a university in the affiliation indicates academic status.’,’discipline’:’The
witness’ discipline if they are an academic, for instance, economics or psychology. If there is
more than one discipline, include them all.’,’university_affiliation’:’If the witness has a
university affiliation, state it here. Otherwise leave empty.’,’full_committee’:’The full committee
 of the hearing. If more than one full committee exists, list them all separated by a colon (:).’,’
sub_committee’:’The subcommittee of the hearing.’,’sub_committee_count’:’The total number of
subcommittees included in the hearing.’,’title_description’:’The title of the hearing.’,’chamber’:’
House of Representatives, Senate, or Joint.’}},]. Text: {testo}"}
    ]
```

Appendix B: Top 30 economists' congressional witness, 1969-2023

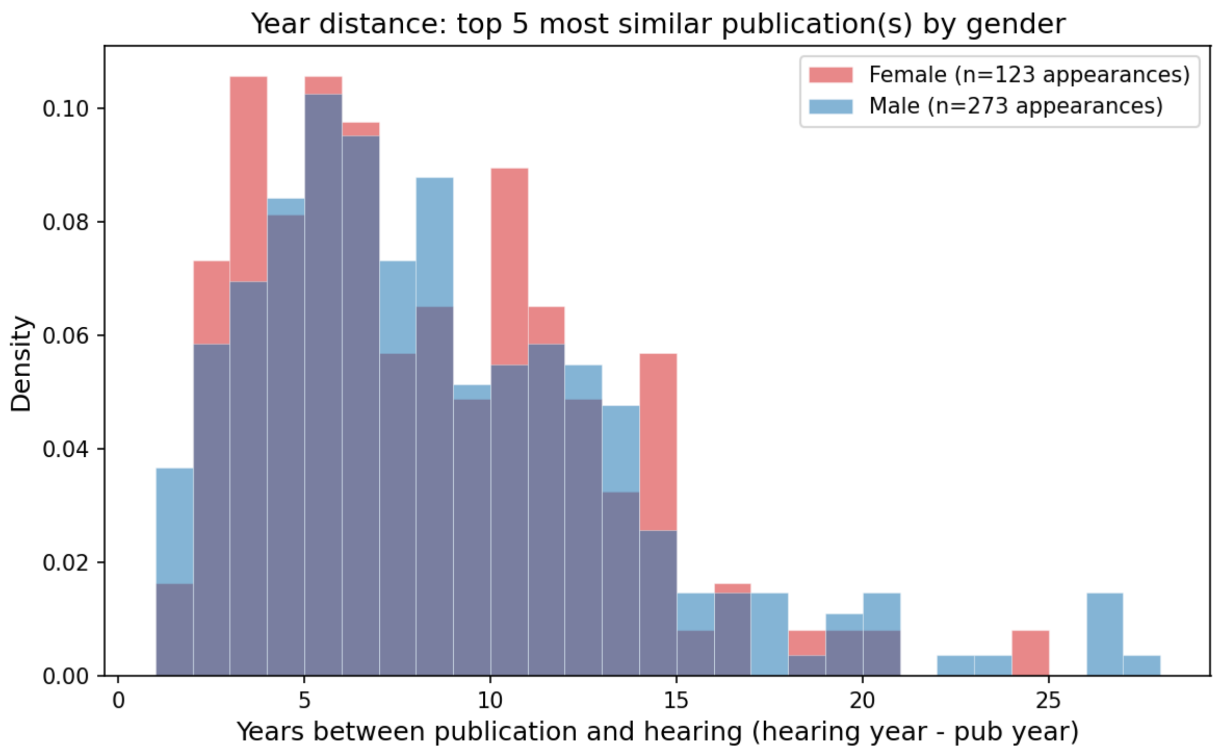
Witness	Appearances	First year	Last year	Share (%)
Robert Eisner	58	1969	1997	1.09
Allan Meltzer	46	1972	2015	0.86
Martin Feldstein	44	1972	2015	0.83
Walter Heller	42	1969	1986	0.79
Jeffrey Sachs	41	1986	2010	0.77
Lester Thurow	38	1970	1999	0.71
Simon Johnson	34	2008	2016	0.64
Murray Weidenbaum	33	1972	1999	0.62
John Galbraith	31	1969	1992	0.58
John Taylor	29	1992	2023	0.55
Karen Davis	28	1981	1992	0.53
Lawrence Klein	28	1969	1989	0.53
Uwe Reinhardt	28	1975	2013	0.53
Lawrence White	27	1974	2014	0.51
Hendrik Houthakker	26	1971	1991	0.49
Alan Blinder	26	1980	2017	0.49
Gary W. Williams	24	1991	2023	0.45
Alan Auerbach	24	1981	2011	0.45
Benjamin Friedman	24	1976	2007	0.45
James Tobin	24	1971	1993	0.45
Dale Jorgenson	23	1971	2012	0.43
Rudiger Dornbusch	23	1977	1995	0.43
Richard Cooper	23	1969	2005	0.43
Alfred Kahn	22	1969	2003	0.41
Edward Kane	22	1975	2014	0.41
Paul Samuelson	21	1970	1993	0.39
Paul Roberts	20	1970	1987	0.38
Laurence Kotlikoff	19	1989	2015	0.36
Walter Adams	19	1969	1990	0.36
Francis Thomas Juster	19	1973	1994	0.36

Appendix C: University affiliation of economists' congressional witness, 1969-2023

Witness University Affiliation	# of Testimonies	Share (%)	Witness University Affiliation	# of Witnesses	Share (%)
Harvard University	512	9.63	Harvard University	112	4.84
Mit	239	4.49	University Of California Berkeley	57	2.46
University Of Pennsylvania	172	3.23	University Of Pennsylvania	55	2.38
University Of Michigan	153	2.88	George Mason University	53	2.29
Stanford University	153	2.88	University Of Maryland	52	2.25
George Mason University	134	2.52	Stanford University	51	2.20
Yale University	129	2.43	University Of Chicago	48	2.07
University Of Maryland	126	2.37	University Of Michigan	46	1.99
Princeton University	123	2.31	Mit	42	1.81
Columbia University	117	2.20	University Of Wisconsin Madison	41	1.77
University Of California Berkeley	113	2.12	Yale University	41	1.77
University Of Chicago	105	1.97	Columbia University	39	1.68
Georgetown University	101	1.90	Cornell University	36	1.56
University Of Minnesota	98	1.84	Princeton University	35	1.51
Carnegie-Mellon University	98	1.84	Georgetown University	35	1.51
Northwestern University	96	1.80	University Of Minnesota	32	1.38
Cornell University	93	1.75	Iowa State University	27	1.17
Johns Hopkins University	79	1.49	George Washington University	27	1.17
University Of Wisconsin Madison	66	1.24	Nyu	26	1.12
Michigan State University	65	1.22	Johns Hopkins University	25	1.08
George Washington University	63	1.18	Rutgers University	25	1.08
Iowa State University	62	1.17	Duke University	25	1.08
Nyu	57	1.07	University Of Illinois	23	0.99
Texas A&M University	53	1.00	Indiana University	23	0.99
University Of Texas	51	0.96	Michigan State University	23	0.99
Duke University	50	0.94	University Of California	22	0.95
Rutgers University	49	0.92	University Of Missouri	22	0.95
University Of Virginia	48	0.90	Carnegie-Mellon University	21	0.91
Ohio State University	45	0.85	University Of California Davis	20	0.86
University Of Illinois	44	0.83	Ohio State University	20	0.86

The left column is total number of testimonies per university, and the right column is number of unique economists per university.

Appendix D: Years Between most Similar Publications and Hearing



Notes: For all hearings between 2017-2023, the witness' testimony has been extracted from the transcript, encoded using LogicMill, and the cosine similarity has been calculated against the witness' stock of publications. For each witness, the time between publication and testimony for the top-5 most similar publications are plotted here.

Appendix E: Committee Groupings and their Corresponding Committees

Committee Type	Committees
Agriculture	Committee on Agriculture; Committee on Agriculture, Nutrition, and Forestry; Committee on Agriculture and Forestry; Committee on Merchant Marine and Fisheries
Banking and Finance	Committee on Finance; Committee on Financial Services; Committee on Banking, Housing, and Urban Affairs; Committee on Banking, Currency, and Housing; Committee on Banking, Finance, and Urban Affairs; Committee on Banking and Financial Services; Committee on Banking and Currency; House Committee on Banking and Financial Services
Budget and Appropriations	Committee on Ways and Means; Committee on the Budget; Committee on Appropriations; Committee on Budget; Study Committee on Budget Control
Economy	Committee on Economic; Committee on Small Business and Entrepreneurship; Joint Economic Committee; Committee on Small Business; Committee on Small Business, Select; Special Committee to Study Problems of American Small Business; Economic Committee; Committee on Economic Report
Energy and Commerce	Committee on Interstate and Foreign Commerce; Committee on Energy and Commerce; Committee on Commerce; Select Committee on the Climate Crisis; Committee on Energy Independence and Global Warming, Select; Committee on Atomic Energy
Infrastructure	Committee on Public Works; Committee on Commerce, Science, and Transportation; Committee on Transportation and Infrastructure; Committee on Environment and Public Works; Committee on Public Works and Transportation; Committee on Post Office and Civil Service
Internal	Committee on Rules and Administration; Committee on House Administration; Committee on Rules; Select Committee on the Modernization of Congress; Committee on Standards and Conduct, Select; Committee on the Organization of Congress
Judiciary	Committee on the Judiciary

Continued on next page

Committee Type	Committees
Labor and Welfare	Committee on Children, Youth, and Families, Select; Committee on Labor and Public Welfare; Committee on Nutrition and Human Needs, Select; Committee on Health, Education, Labor, and Pensions; Special Committee on Aging; Committee on Education and the Workforce; Committee on Education and Labor; Committee on Veterans' Affairs; Committee on Labor and Human Resources; Committee on Equal Educational Opportunity, Select; Committee on Economic and Educational Opportunities; Committee on Labor-Management Relations; Joint Select Committee on Solvency of Multiemployer Pension Plans; Committee on Narcotics Abuse and Control, Select; Committee on Aging, Select; Committee on Hunger, Select; Committee on Human Resources; Committee on Crime, Select; Special Committee on Unemployment Problems
Natural Resources	Committee on Energy and Natural Resources; Committee on Natural Resources; Committee on Interior and Insular Affairs; Committee on Public Lands; Committee on Resources; Committee on National Water Resources, Select
Other	House Commissions and Temporary Committees; Committee on Indian Affairs; Committee on the District of Columbia; Commissions and Temporary Committees; Committee to Investigate the Use of Chemicals in Food Products, Select; Committee to Investigate the Use of Chemicals in Foods and Cosmetics, Select; Committee on Population, Select; Committee on Un-American Activities; Committee on Indian Affairs, Select; Committee on Democratic Policy; Committee on Washington Metropolitan Problems
Oversight	Committee on Governmental Affairs; Committee on Government Operations; Committee on Oversight and Government Reform; Committee on Oversight and Reform; Committee on Oversight and Accountability; Committee on Government Reform and Oversight; Committee on Government Reform; Committee on Oversight; House Committee on Government Reform and Oversight; Committee on Expenditures in Executive Departments
Science	Committee on Science and Technology; Committee on Science, Space, and Technology; Committee on Science; Committee on Science and Astronautics; Committee on Aeronautical and Space Sciences; Ad Hoc Select Committee on the Outer Continental Shelf

Continued on next page

Committee Type	Committees
Security and Foreign Affairs	Committee on Homeland Security; Committee on Foreign Relations; Committee on Homeland Security and Governmental Affairs; Committee on Foreign Affairs; Commission on Security and Cooperation in Europe; Committee on Armed Services; Commission on Security & Cooperation in Europe; Committee on International Relations; Committee on National Security; Committee to Investigate Communist Aggression and the Forced Incorporation of the Baltic States into the Soviet Union, Select; Committee on Internal Security; Select Committee on Intelligence; Congressional-Executive Commission on China; Committee on Naval Affairs

Appendix F: Conditional Logit Stratified on Hearing

	(1)	(2)	(3)	(4)	(5)	(6)
Female	-0.01	-0.13^{*}	-0.17^{**}	-0.23^{***}	0.37^{*}	0.41^{**}
	(0.06)	(0.07)	(0.07)	(0.07)	(0.15)	(0.16)
Log Citations/Pubs		0.42^{***}	0.40^{***}	0.41^{***}	0.40^{***}	0.40^{***}
		(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Log Coauthors		-1.20^{***}	-1.19^{***}	-1.19^{***}	-1.19^{***}	-1.19^{***}
		(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Seniority		0.02^{***}	0.02^{***}	0.02^{***}	0.02^{***}	0.02^{***}
		(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Log Top 5 Pubs (stock)		0.14^{***}	0.13^{***}	0.13^{***}	0.13^{***}	0.12^{***}
		(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Log policy mentions (stock) t1		-0.16^{***}	-0.16^{***}	-0.15^{***}	-0.15^{***}	-0.15^{***}
		(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Log media mentions (stock) t1		0.12^{***}	0.12^{***}	0.14^{***}	0.13^{***}	0.14^{***}
		(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Top 10 University		0.42^{***}	0.47^{***}	0.47^{***}	0.47^{***}	0.47^{***}
		(0.04)	(0.04)	(0.04)	(0.04)	(0.04)
Log Driving Time			-0.18^{***}	-0.18^{***}	-0.15^{***}	-0.15^{***}
			(0.02)	(0.02)	(0.02)	(0.02)
Virtual						
Female x Virtual				0.91^{***}		-0.07
				(0.23)		(0.52)
Female x Log Driving Time					-0.29^{***}	-0.34^{***}
					(0.08)	(0.08)
Virtual x Log Driving Time						-0.21
						(0.14)
Female x Virtual x Log Driving Time						0.52^{*}
						(0.23)
N	48062	48062	48062	48062	48062	48062
Events	3967	3967	3967	3967	3967	3967
Clusters	2955	2955	2955	2955	2955	2955
Log-likelihood	-11354.73	-7645.49	-7613.51	-7607.60	-7606.61	-7597.96

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$; $^{\cdot}p < 0.1$

Coefficients are reported as log-odds. Standard errors are in parentheses. The conditional logit model stratifies by hearing, comparing each actual witness against the pool of eligible at-risk economists for that hearing. Standard errors are clustered at the hearing level. *** p < 0.001; ** p < 0.01; * p < 0.05; † p < 0.10.

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